When Care Supply Becomes Care Demand: How Older Couples Adjust to Caregiver Decline

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Abstract

A healthy spouse insures older people against long-term care needs, but the caregiving spouse is aging too. Using SHARE, I follow couples in which one partner cares for the other as the caregiver develops new limitations in activities of daily living. Multiple new limitations make the care relationship about 20 percentage points more likely to end. Help from children and home services rises, but much of it goes to the caregiver. Recipients whose caregiver develops multiple limitations are about twice as likely to report that help does not consistently meet their needs: more than one in two, compared with about one in four. More support around a couple does not mean that lost partner care has been replaced.

Keywords: spousal caregiving; long-term care; family insurance; informal care; unmet care needs; SHARE
JEL codes: J14, D13, J12, I18, I11

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1 Introduction

A healthy spouse is one of the most important forms of insurance against long-term care needs in old age (Lakdawalla and Philipson, 2002; Fadlon et al., 2026). Across 23 OECD countries, 70 percent of long-term care recipients receive care at home (OECD, 2025), and a partner's help is often what makes this possible. Yet this insurance has an unusual feature: the person who provides it is exposed to the very risk it covers. In the Survey of Health, Ageing and Retirement in Europe (SHARE), about one in ten partners who provide personal care to a spouse report a new limitation in activities of daily living (ADL) by the next interview. When that happens, the household loses a source of care and gains a second person who needs it. This paper asks two questions about what follows: who receives the new help, and is it enough for the partner who was already dependent?

Both questions are hard to answer with household-level measures of support, which do not reveal whom assistance reaches or whether it is sufficient for each partner. Consider a couple in which one partner helps the other dress and bathe. When the helper develops mobility difficulties and an adult child or a home service becomes involved, the new help may take over the tasks the partner used to perform, or it may assist the helper. Both arrangements raise household support, but only the first replaces the care that the dependent partner has lost. Treating more household support as replacement would then overstate the help available to the person who was already dependent.

This paper studies how care is reorganized when the caregiving partner develops functional limitations. Using SHARE, I identify couples in which a partner without ADL limitations provides personal care to a partner with at least one limitation, and I keep these roles fixed as both people's health and support change. The design observes who was providing care before the decline, follows outside help from adult children and home services to each partner separately, and records whether the original recipient reports that help meets needs. I compare caregivers who develop one or multiple new ADL limitations with caregivers who remain free of limitations, using overlap weighting and regression adjustment for baseline differences. The estimates are adjusted associations rather than causal effects of health shocks, but they show how care arrangements change around the caregiver's decline.

Three findings emerge. First, partner-care exit rises sharply only with multiple new limitations. One new limitation does not detectably change whether partner care continues, while multiple new limitations make the original relationship about 20 percentage points more likely to end, raising the exit probability from about one-half to about two-thirds. Second, outside help arrives as a combination of family and formal support, and much of it goes to the caregiver. Households become more likely to receive help from children and

home services together, while home services without help from children do not increase. The caregiver's probability of starting to receive help from children roughly doubles, and the share of caregivers who start receiving home personal care rises from about one in a hundred to about one in nine. Most of the additional entry by the original recipient occurs jointly with entry by the caregiver. Third, the original recipient's needs are less often met. When the caregiver develops multiple limitations, the share of recipients with continuing ADL limitations who report that help does not consistently meet their needs rises from about one in four to more than one in two. Almost all of this increase comes from recipients who receive help that is not enough, rather than from recipients who receive none, and it occurs in the same households that more often receive both forms of outside support. Put concretely, among 100 couples in which the caregiver develops multiple limitations, about 20 more care relationships end and about 23 more households receive help from both children and home services than among comparable couples in which the caregiver stays free of limitations; among 100 recipients who still need help, about 28 more report that it is not enough.

Two features of the adjustment are consistent with more outside support coexisting with less adequate help. Part of the new support addresses the caregiver's own needs rather than replacing the care the recipient has lost. And outside help differs in kind from care by a co-resident partner, who is present throughout the day and can help whenever a need arises; children who live elsewhere and scheduled home services help at particular times. The timing of the adjustment points in the same direction: the point estimates show partner care weakening before the caregiver's limitation is recorded, whereas outside support rises afterwards. Responses also differ across caregiving roles and care systems. With multiple new limitations, the exit difference is 33 percentage points for male caregivers and 10 for female caregivers, without a correspondingly larger outside response. In Central and Eastern Europe, where home personal care is less available, caregivers who develop multiple limitations rarely start receiving it, partner care is not more likely to end, and help from children to the caregiver rises instead.

This paper contributes first to research on the family as insurance against long-term care needs. A healthy partner reduces demand for institutional care (Lakdawalla and Philipson, 2002), and family arrangements shape the effects of long-term care policy and the demand for long-term care insurance (Barczyk and Kredler, 2018; Ko, 2022; Mommaerts, 2025; Coe et al., 2023). In Barczyk and Kredler (2018), care between spouses is provided automatically while the caregiver is healthy, and Capatina et al. (2024) allow both spouses to need care, with informal care less likely when the spouse is also disabled. I complement these models by following an observed care relationship through the caregiver's functional decline and documenting changes in partner care, the destination of outside assistance, and the original

recipient's help adequacy. Closest to this paper, Fadlon et al. (2026) use linked Medicare records to show that a spouse's sudden incapacitation increases the partner's probability of a nursing-home stay. I show that the protection a partner provides depends on that partner's own functional capacity and weakens sharply once the caregiver has multiple limitations. Rather than reliance on institutional care, I study the care relationship at home and document a second consequence of the provider's decline: the provider becomes a user of help. Related work examines how families respond to a member's health deterioration, including labor-supply and caregiving responses (Fadlon and Nielsen, 2021; Macchioni Giaquinto et al., 2022), health spillovers (Arteaga et al., 2025), and institutional care needs after bereavement (Schouwenaar et al., 2026), and documents the health costs of spousal and intensive caregiving (Bom et al., 2019; Stöckel and Bom, 2022). Bom et al. (2019) also note that older spousal caregivers are underrepresented in their sample of people aged 45 to 65; this paper studies such caregivers directly and follows what happens to the person they care for.

Second, the paper contributes to research on how informal and formal care relate to each other. Studies examine whether help from relatives substitutes for home services and institutional care (Van Houtven and Norton, 2004; Charles and Sevak, 2005; Bonsang, 2009; Bergeot and Tenand, 2023), and how expanding services changes family care and adult children's employment (Løken et al., 2017; Hollingsworth et al., 2022; Shen, 2024). This literature studies substitution between care sources and responses to changes in service provision. When the existing caregiver also develops care needs, substitution also depends on whom new assistance reaches; I add evidence on this question. I find that home services enter alongside help from children rather than in its place, and that a large part of the new help goes to the former caregiver. The results also extend evidence on adult children's caregiving. Barszczewski et al. (2025) show that women increase caregiving as their parents' health worsens and document the associated labor-supply costs. In couples where one parent already cares for the other, help from children rises when the caregiving parent declines, and much of it goes to that parent.

Third, the paper contributes to the measurement of unmet care needs. Research documents unmet needs across European countries and socioeconomic groups (Albuquerque, 2022; Szenkurök et al., 2025) and changes in home-care use and unmet needs during the COVID-19 pandemic (Arabadzhyan et al., 2025). I show that unmet needs can rise within households that receive more outside help, and that the shortfall mainly concerns help that is received but insufficient. Measures that record only whether a household receives help would miss this shortfall.

Section 2 presents the conceptual framework, data, and empirical strategy. Section 3 examines partner-care exit and the household's outside support. Section 4 identifies who

receives the new support, relates it to the recipient's reported help adequacy, and examines differences by gender and region. Section 5 examines the persistence and timing of these changes and presents robustness checks. Section 6 concludes.

2 Framework, Data, and Empirical Strategy

2.1 Conceptual framework

An existing partner-care relationship connects one person's need for assistance to the other person's ability to provide it. Care requires time, effort, and physical capacity, making the caregiver's health part of the household's productive resources (Becker, 1965; Grossman, 1972). When the caregiver develops functional difficulties, providing care may become harder and the caregiver may begin to need assistance.

How far these two changes go depends on the extent of functional difficulty. A single difficulty may create a need for help without preventing a caregiver from assisting the partner. Multiple difficulties may interfere with a wider range of caregiving tasks. Comparing these groups shows how outside support and partner-care exit differ across degrees of functional difficulty.

Children and home services face different constraints. Adult children allocate time across parents, employment, and other responsibilities, while home services depend on availability, affordability, and eligibility. The literature on family care and service provision shows that these sources can substitute for one another and that their relationship depends on care needs (Barczyk and Kredler, 2018; Van Houtven and Norton, 2004; Bonsang, 2009). Separating help from children from home services allows the analysis to establish which channels become involved and whom they reach.

Outside support can respond to either change: the loss of care for the original recipient or the caregiver's new need. As in the example in the Introduction, help from children or home services can reach the caregiver, the original recipient, or both, while producing the same household indicator of support receipt. Following both partners separately identifies whose needs the outside response addresses. Receipt shows whether help is present, not whether it is enough. Whether the original recipient's lost care has been replaced therefore also depends on that person's report of whether help meets needs.

The empirical analysis follows this logic in three steps: establish how partner care changes when the caregiver develops limitations; measure the response of outside support around the couple; and identify new receipt by each partner alongside the original recipient's reported help adequacy. Regional and gender comparisons examine how these responses vary across

care environments and caregiving roles.

2.2 Data and measurement

I use the Survey of Health, Ageing and Retirement in Europe (SHARE), which follows people aged 50 and older and their partners (Börsch-Supan and Jürges, 2005; Börsch-Supan et al., 2013). Release 9.0.0 provides regular interviews from waves 1, 2, and 4–9; wave 3 is retrospective and is excluded. The sample covers European countries, excluding Israel, and includes married and cohabiting partners. Each observation links a couple's baseline interview to the next regular survey wave in which their country participates. A couple can contribute more than one observation. Wave-specific methods and participation and attrition patterns are documented in Bergmann et al. (2024) and Bergmann et al. (2025), respectively.

Baseline roles and functional health. The original caregiver (G) has no limitation in activities of daily living (ADL) and provides personal care to a partner with at least one limitation, the original recipient (R). These labels remain fixed through follow-up, including when the caregiver begins receiving assistance or the recipient recovers.

The ADL measure covers six activities: dressing, walking across a room, bathing, eating, getting in or out of bed, and toileting. The main comparison distinguishes caregivers who develop at least one limitation from those who continue to report none. I then separate one new limitation from multiple new limitations, defined as two or more. ADL difficulties are directly relevant to the physical tasks involved in personal care, whereas a medical diagnosis need not impair the ability to provide that care.

Instrumental activities of daily living (IADL) capture broader baseline difficulties. I reconstruct a common seven-item count covering map use, meal preparation, shopping, telephone use, medication, housework or gardening, and money management. Two items added from wave 6 are excluded to maintain the same scale across waves. Earlier caregiver ADL limitations are also recorded, including episodes followed by recovery before baseline.

Partner-care continuation. Either partner can establish the baseline relationship: the caregiver reports providing personal care to the partner, or the recipient reports receiving it from that person. To assess continuation, I follow the same reporter across interviews. When the recipient confirms baseline care, I use the recipient's report at both dates; when only the caregiver confirms it, I use the caregiver's report. Partner-care exit equals one when that reporter no longer reports the relationship at follow-up. Unavailable answers remain

missing.

Two sources of outside support. Help from children includes help from children or stepchildren living elsewhere and regular personal care from co-resident children or stepchildren. Help from children living elsewhere can involve personal care, practical household assistance, or paperwork. Sons-in-law, daughters-in-law, and grandchildren are excluded from this channel.

Home services comprise home personal care, domestic assistance, and delivered meals. The channel is defined by receipt rather than payment because SHARE does not identify a common payer across countries.[1] I examine both any home-service receipt and its three components.

Household receipt and individual entry. Household receipt equals one when either partner receives a channel at follow-up. A zero requires valid information establishing that neither receives it. With both channels measured, the couple belongs to one of four categories: help from children only, home services only, both, or neither.

Individual entry identifies who starts receiving assistance. For channel k and original role $j \in \{G, R\}$, let S^k_{ji0} and S^k_{ji1} denote receipt at baseline and follow-up. Entry is

$$N^k_{ji} = \mathbf{1}\{S^k_{ji0} = 0,\ S^k_{ji1} = 1\}. \tag{1}$$

Both partners must have valid support information at both dates. Baseline users remain in the sample and have entry equal to zero. The difference $N^k_{Gi} - N^k_{Ri}$ compares entry between the original roles. I also distinguish entry by the caregiver only, the recipient only, both partners, and neither.

Question wording determines whether receipt can be assigned to a person. In waves 1, 2, 4, and 5, questions about help from people living elsewhere refer jointly to the respondent and partner. A positive answer identifies household receipt but not its individual destination. A valid joint baseline zero can establish that neither partner received help from children living elsewhere; personal care from co-resident children is added separately. Individual child-help entry requires individually worded follow-up questions. Home-service receipt comes from each person's health-care interview.

These differences explain why household and individual outcomes use different samples. Wave 4 lacks home-service questions, and wave 7 respondents routed to the life-history

[1] In waves 1 and 2, home nursing and personal care are combined in one question. Home personal care includes that item in these waves and the corresponding personal-care item in later waves. Institutional residence is measured separately.

questionnaire were not asked the relevant support questions. Unasked questions are treated as missing. Appendix A provides the wave-specific rules, and Appendix Table A.1 maps the outcomes to the original questions.

Help adequacy. For recipients who still report ADL limitations, I also examine whether help meets their needs. SHARE asks whether anyone helps with difficult activities and, when help is received, whether it meets needs all the time, usually, sometimes, or hardly ever. Help is classified as *not consistently sufficient* when no activity help is received or received help meets needs less than all the time. A stricter measure counts no help or help that meets needs only sometimes or hardly ever. These questions concern assistance with difficult activities more broadly than personal care alone.

2.3 Sample construction and follow-up

The baseline sample contains 6,359 observations beginning with an existing personal-care relationship: a partner without ADL limitations is providing care to a partner with limitations. Follow-up is scheduled using the country survey calendar, whether or not the couple returns. Inclusion does not require care to end or the recipient to remain limited.

To construct this sample, I link partners using reciprocal identifiers and require a common household and country, completed baseline interviews, and valid ADL information for both people. Reports of personal-care provision or receipt identify 8,018 European couple-wave records meeting the baseline role definitions. I exclude 1,365 wave 9 records without a later wave in this release, 279 records preceded by an observed episode in which both partners had ADL limitations, and 15 without a subsequent country survey wave. The resulting 6,359 observations retain caregivers with earlier limitations who have recovered by baseline; their observed ADL history enters the adjustment variables.

Table 1 shows what can be established about caregiver health at the scheduled follow-up. Health change is classified in 4,722 observations, including 477 with new limitations. Caregiver death and unavailable health information are recorded separately.

The analysis samples additionally require the information needed for each outcome (Table 2). Household support requires both partners to be alive, living together in the community, reciprocally linked, and observed with valid caregiver health and support information. Individual entry additionally requires baseline support information and personal attribution at follow-up. The primary exit analysis excludes follow-ups in wave 7 because the life-history questionnaire did not ask the relevant care questions. Help adequacy requires the original recipient to remain ADL-limited and provide valid help responses.

Table 1: Caregiver health at the scheduled follow-up

Follow-up status	Observations
Valid caregiver ADL information	4,722
No new limitation	4,245
One new limitation	280
Multiple new limitations	197
Caregiver death	246
No completed caregiver main interview	1,375
Completed interview, ADL information missing	16
Baseline sample	6,359

Notes: Each observation links a baseline couple to its next scheduled country survey wave. The three indented rows partition observations with valid caregiver ADL information. Multiple new limitations means two or more.

Table 2: Analysis samples by caregiver functional change

		New caregiver ADL limitations		
Outcome sample	Total	None	One	Multiple
Partner-care exit	2,581	2,341	136	104
Household support	2,537	2,317	131	89
All three home-service types	2,533	2,315	130	88
Entry into help from children	1,528	1,416	68	44
Entry into home services	1,890	1,753	86	51
Entry into home personal care	1,890	1,753	86	51
Recipient help adequacy	1,423	1,319	63	41
Household support and adequacy	1,381	1,278	63	40

Notes: Samples reflect the information required for each outcome and are not successive selection steps. All caregivers have zero ADL limitations at baseline. Multiple means two or more limitations at follow-up. Within each channel, the two role-entry outcomes use the same observations. Home personal-care entry uses the home-service entry sample. Adequacy requires continuing recipient ADL limitations and valid help responses.

Community follow-up also differs across health groups. Among observations with new caregiver limitations, 34.4 percent fail the community-couple conditions, compared with 25.9 percent without new limitations. Recipient death is recorded in 132 of 477 observations with new caregiver limitations (27.7 percent) and 798 of 4,245 without (18.8 percent). Appendix D reports death, institutional residence, and other follow-up states.

2.4 Estimation

2.4.1 Overlap weighting and outcome regression

The main estimator combines overlap weighting with weighted regression adjustment. A logit model predicts the caregiver's follow-up health state from baseline characteristics and supplies overlap weights; a weighted linear probability model then estimates the difference in each care outcome between the health groups. Caregivers who develop limitations differ at baseline from those who do not: in the household-support sample, their ages average 73.8 and 69.0 years, and their IADL counts average 0.48 and 0.17. Weighting balances the characteristics included in the logit model. The outcome regression adjusts for differences that remain after weighting, in particular country composition and recipient characteristics outside the propensity model (Table 3).

Propensity score and overlap weights. Let $D_i = 1$ when the caregiver develops at least one ADL limitation and $D_i = 0$ when the count remains zero. I first fit a logit model for D_i in each outcome sample. Its predictors are region, caregiver age and gender,[2] caregiver seven-item IADL count and earlier ADL history, recipient ADL count, confirmation of baseline care by both partners, baseline wave as a linear term, and years between survey waves. Health, demographic, and care-reporting controls are measured at baseline. Earlier ADL history distinguishes a previous limitation, earlier valid interviews without limitations, and unavailable history. Survey spacing is calculated from country-wave median interview dates.

Region has three predefined categories: Northern and Western, Southern, and Central and Eastern Europe.[3] Some outcome samples contain countries without new caregiver limitations. Country indicators would perfectly predict health status in these countries and cause their observations to be dropped from the logit fit. I therefore use region in the propensity model and country effects in the outcome regression.

I then convert the estimated propensity scores into overlap weights:

$$\widehat{e}_i = \widehat{\Pr}(D_i = 1 \mid Z_i), \qquad w_i = D_i(1 - \widehat{e}_i) + (1 - D_i)\widehat{e}_i. \tag{2}$$

An observation with new limitations receives weight $1 - \widehat{e}_i$, while an observation without

[2]Gender is taken from the SHARE coverscreen, which records each respondent as male or female. SHARE does not provide a separate measure of gender identity.

[3]Northern and Western Europe comprises Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Luxembourg, the Netherlands, Sweden, and Switzerland. Southern Europe comprises Cyprus, Greece, Italy, Malta, Portugal, and Spain. Central and Eastern Europe comprises Bulgaria, Croatia, the Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Slovakia, and Slovenia. Ireland contributes no eligible observations after the scheduled-follow-up restriction.

new limitations receives weight $\widehat{e}_i$. Couples whose baseline characteristics make either health state plausible receive greater influence in the comparison (Rosenbaum and Rubin, 1983; Li et al., 2018). The comparison therefore refers to this overlap-weighted population rather than to all caregivers in either health group. Each outcome has its own estimation sample and fitted weights. No observations are trimmed.

Weighted outcome regression. For a binary care outcome Y_i, I estimate the following linear probability model by weighted least squares, using w_i as the observation weight:

$$Y_i = \alpha + \tau D_i + Z_i'\beta + \gamma_{c(i)} + \rho_1 A_{Ri0} + \rho_2 I_{Ri0} + u_i, \tag{3}$$

where Z_i contains the propensity-model covariates, and A_{Ri0} and I_{Ri0} are recipient baseline age and IADL. Country indicators, denoted by $\gamma_{c(i)}$, allow a separate intercept for each country and absorb the regional indicators. Repeated observations from the same couple are handled through the couple-level bootstrap described below.

The coefficient τ is the adjusted difference in the probability of the outcome between caregivers with and without new limitations, holding country and the other regressors fixed. Tables report 100τ in percentage points, together with the observed comparison-group mean under the same weights. To illustrate the magnitude of selected results, I add the adjusted difference to this weighted comparison-group rate. The sum is a model-implied probability with new limitations, standardized to the weighted comparison group's baseline characteristics, rather than an observed rate in either group; because the outcome regression also adjusts for country and recipient characteristics outside the propensity model, it need not equal the average predicted probability in the overlap-weighted population. For partner-care exit, 100τ measures the adjusted difference in the probability that the baseline relationship is no longer reported. Section 5 reports alternative specifications, including adjustment for the baseline configuration of help from children and home services.

Severity. To compare degrees of functional difficulty, I replace D_i with separate indicators for one new limitation, D_i^1, and multiple new limitations, D_i^M:

$$Y_i = \alpha + \tau_1 D_i^1 + \tau_M D_i^M + Z_i'\beta + \gamma_{c(i)} + \rho_1 A_{Ri0} + \rho_2 I_{Ri0} + u_i. \tag{4}$$

Zero limitations is the omitted category. The regression retains the weights w_i estimated for any new limitation versus none, so both severity groups are compared with the same weighted comparison group. The coefficients τ_1 and τ_M measure each group's adjusted difference from zero limitations, and $\tau_M - \tau_1$ gives the multiple-minus-one contrast. Gender and regional

comparisons instead re-estimate weights for multiple new limitations versus none, excluding one-limitation observations (Section 4.3). Section 5 also reports this binary comparison for the main outcomes.

Original roles. Individual-entry models add both partners' baseline receipt in the relevant channel to the propensity and outcome models. Within each channel, I estimate equation (3) separately for caregiver entry N_{Gi}^k and recipient entry N_{Ri}^k, using identical observations, weights, and regressors. To compare the two roles, I also estimate the same equation with their difference $N_{Gi}^k - N_{Ri}^k$ as the dependent variable. Because the model is linear and the specification is unchanged, its coefficient equals the caregiver coefficient minus the recipient coefficient. The couple-level bootstrap re-estimates this difference regression in each draw, so its standard error reflects the correlation between the two partners' outcomes within a couple. The same approach applies to the four mutually exclusive entry destinations: caregiver only, recipient only, both partners, and neither.

Recipient help adequacy. Adequacy models use the same estimator and add the recipient's baseline help adequacy, classified as adequate, inadequate, or unavailable, to both the propensity and outcome models. Comparisons that combine household support and recipient adequacy use their common sample. Appendix Table D.1 also reports estimates without the baseline adequacy adjustment.

2.4.2 Balance and inference

Table 3 reports baseline balance in the household-support sample. Overlap weights exactly balance the means of covariates included in the logit model, so the caregiver age and IADL differences are removed. Recipient age and IADL, which are outside the propensity model, also show small weighted differences. Country composition and baseline support remain less closely aligned; country effects enter the outcome regression, and the baseline-support specification examines the latter difference.

The household comparison has effective sample sizes of 1,566 observations without new limitations and 216 with new limitations. Appendix B reports these diagnostics for the individual-entry samples, together with alternative specifications.

Inference repeats the full estimation procedure in a couple-level bootstrap. Each draw resamples couples with all their observations, refits the propensity model, recalculates overlap weights, and refits the weighted outcome regression. This preserves dependence across repeated observations for the same couple and incorporates estimation of the weights. The primary models, timing comparison, and reporting checks use 999 draws; supplementary

Table 3: Baseline balance before and after overlap weighting

	Standardized mean difference	
	Unweighted	Weighted
Caregiver age	0.52	0.00
Caregiver IADL	0.42	0.00
Recipient ADL	0.05	0.00
Years between survey waves	−0.08	0.00
Both partners confirm baseline care	−0.12	0.00
Recipient age†	0.42	−0.01
Recipient IADL†	0.03	−0.08
Baseline help from children†	0.15	0.06
Baseline home services†	0.07	−0.12
Country composition: maximum†	0.22	0.23
Observations	2,537	
Caregivers with any new limitation	220	

Notes: Standardized differences use the square root of the average unweighted group variance. The country row reports the maximum absolute difference across country indicators. † marks variables outside the propensity model. Included logit terms balance exactly under overlap weighting; region, baseline wave, caregiver gender, and caregiver ADL history are also included. Baseline support is evaluated on available records using the same weights: 2,494 observations for help from children and 1,869 for home services.

analyses use 399 where indicated. An estimate is reported with bootstrap inference when at least 80 percent of draws fit both models without dropping observations. Significance stars use unadjusted two-sided tests based on the bootstrap standard errors unless noted otherwise. Gender and regional comparisons involve many interaction tests, so I report Benjamini–Hochberg adjusted q-values for them (Benjamini and Hochberg, 1995); Appendix E gives the test families.

The estimates describe adjusted associations between caregiver functional change and care arrangements. Health and support are recorded at the same interview, and Section 5.2 examines their timing. Household-support and role-entry results concern couples living together in the community; death, institutional residence, and other follow-up states are examined separately.

3 Partner-Care Exit and the Response of Outside Support

Caregiver functional decline is associated with two changes in the household: the original care relationship becomes more likely to end, and outside support becomes more common. The

extent of functional difficulty distinguishes how these responses combine. Table 4 presents them together, comparing any new limitation, one limitation, and multiple limitations with none. Panel A reports partner-care exit; Panels B and C report household receipt of outside support by channel and by household configuration.

Table 4: Partner-care exit and household outside support

	Rate, no new limitation (%)	Any new limitation	One new limitation	Multiple new limitations
A. Original partner-care relationship				
Partner-care exit	47.79	6.75**	−2.86	20.42***
		(3.25)	(4.18)	(4.65)
Observations		2,581		
Caregivers with new limitations		240	136	104
B. Household receipt by channel				
Help from children	31.51	14.67***	9.18**	23.57***
		(3.57)	(4.56)	(5.24)
Home services	37.70	9.46***	3.02	19.93***
		(3.31)	(4.02)	(5.29)
C. Household configuration (mutually exclusive)				
Both channels	14.75	11.14***	3.69	23.26***
		(2.93)	(3.48)	(5.48)
Children only	16.76	3.52	5.50	0.31
		(2.98)	(3.76)	(4.23)
Services only	22.95	−1.68	−0.66	−3.33
		(2.97)	(3.87)	(4.38)
Neither channel	45.54	−12.99***	−8.52**	−20.25***
		(3.26)	(4.17)	(4.40)
Observations		2,537		
Caregivers with new limitations		220	131	89

Notes: Adjusted percentage-point differences relative to caregivers with no new ADL limitation, with couple-bootstrap standard errors below estimates. The first column reports the overlap-weighted percentage among observations whose caregiver develops no new limitation; adding a coefficient gives a model-implied rate standardized to the weighted comparison group's baseline characteristics. The any-limitation coefficient comes from the binary outcome specification. The one- and multiple-limitation coefficients enter a separate outcome regression jointly, retaining the same any-limitation overlap weights. Each outcome has 999 successful bootstrap draws. Household receipt means that either original partner receives the channel; children and services may assist different people. Panels B and C use the same observations. The two channels in Panel B can overlap. The configurations in Panel C are mutually exclusive, and their coefficients sum to zero within each health comparison, up to rounding. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.1 When does the original relationship end?

Panel A of Table 4 shows that additional partner-care exit is concentrated among caregivers with multiple new limitations. Their exit probability is about 20 percentage points higher than that of caregivers who remain free of limitations, whereas one new limitation shows no detectable increase (−2.9 points, s.e. 4.2). The multiple-minus-one difference is 23.3 points (s.e. 5.9). Standardized to the weighted comparison group, the model-implied exit probability with multiple new limitations is about two-thirds, against about one-half in that group. The overall difference of 6.7 points conceals this gradient: counting any new limitation together would obscure where the original source of personal care is most often lost.

Exit means that the baseline reporter no longer reports the partner-care relationship; it does not by itself mean that the recipient is left without care. Partner care also turns over frequently without new caregiver limitations: the weighted comparison-group exit rate is 47.8 percent. The additional exit is not explained by recipients no longer needing help. In the any-new-limitation comparison, exit accompanied by recipient ADL recovery accounts for only 0.1 of the 6.7 points. Nor does the additional exit mainly leave recipients with no help at all. In waves that observe all sources of personal care, multiple caregiver limitations do not measurably raise the share of recipients who still have ADL limitations but no measured personal-care provider (1.0 points, s.e. 4.9). Appendix D distinguishes these and other follow-up states. Section 4.2 examines whether the help that remains meets the recipient's needs.

The severity gradient is consistent with a threshold in caregiving capacity. An isolated difficulty leaves scope to continue helping, while multiple limitations interfere with a wider range of caregiving tasks. A healthy partner is often viewed as informal insurance against long-term care needs (Lakdawalla and Philipson, 2002; Fadlon et al., 2026). This protection depends on the partner's own functional capacity: it weakens sharply once that capacity is limited in several activities, rather than declining in proportion to each new difficulty. The presence of a partner is therefore an incomplete measure of available care: what matters is that person's capacity to perform the assistance on which the recipient relies. The gradient does not depend on using common weights for both severity groups. Re-estimating the weights for multiple new limitations versus none, excluding one-limitation observations, leaves the exit difference at about 20 percentage points (Section 5).

3.2 How does outside support respond?

Panel B of Table 4 reports receipt of each channel by either partner, and Panel C divides households into mutually exclusive configurations. With one new limitation, household re-

ceipt of help from children is 9.2 percentage points more common (s.e. 4.6), from a weighted comparison-group rate of 31.5 percent, while the home-service estimate is small and imprecise (3.0, s.e. 4.0). The share of households receiving neither channel falls by 8.5 points (Panel C). At one new limitation, the detectable outside response is concentrated in help from children, without a clear change in partner care or in home services.

With multiple new limitations, the response changes form. Help from children is 23.6 points more common and home services 19.9 points more common, and Panel C shows that nearly all of the change appears in households receiving both channels. Joint receipt is 23.3 points more common, about two and a half times the comparison-group rate of 14.8 percent, while receipt of neither channel falls by 20.3 points. Receipt of home services without help from children does not increase at any level of severity (-1.7 points for any new limitation and -3.3 for multiple). Home services thus enter alongside help from children rather than in its place. This pattern is consistent with evidence that help from children and formal home care become less substitutable as care needs grow (Bonsang, 2009). It also has a measurement implication: recording service use alone would miss the accompanying family involvement. A remaining concern is that households already receiving one channel at baseline may be more likely to combine both later. In the multiple-versus-zero comparison on the 1,760 observations with baseline support information, adding the baseline configuration of help from children and home services to both models changes the joint-receipt difference only from 22.1 to 19.5 points (Section 5).

Within home services, the response shifts toward hands-on assistance. For any new limitation, household receipt of home personal care is 10.6 points more common, slightly more than receipt of any home service (9.5 points), while domestic assistance is 8.7 points and delivered meals 4.4 points more common (Appendix Table C.1). A household already receiving domestic assistance can add personal care without becoming a new service user, so part of the service response occurs within existing service arrangements.

Severity therefore distinguishes two patterns. With one new limitation, partner care continues and children become more involved, consistent with family help that supplements care by the partner. With multiple new limitations, partner care is more often lost, and children and home services become involved together. Household receipt cannot show whether the additional help supports the caregiver or replaces care for the original recipient. Section 4 follows each partner separately.

4 Who Receives Outside Support, and Is It Sufficient?

This section asks who receives the additional outside support and whether it meets the original recipient's needs. I first follow each partner's entry into support for any new caregiver limitation. I then focus on multiple new limitations, where Section 3 found additional partner-care exit to be concentrated, to examine recipient help adequacy and differences by gender and region.

4.1 The caregiver becomes a recipient of help

Household support rises, and much of the new support goes to the person who had been providing care. Table 5 follows the two original roles on identical observations within each channel. Compared with caregivers who remain free of limitations, caregivers with any new limitation are 13.4 percentage points more likely to start receiving help from children and 10.0 points more likely to start receiving home services. Against weighted comparison-group entry rates of 13.3 and 8.3 percent, these differences roughly double the caregiver's probability of starting help from children and more than double it for home services. Home personal care shows the change most clearly. The comparison-group entry rate for caregivers is 1.0 percent; the adjusted difference of 10.2 points raises it to about one in nine. The partner who had supplied hands-on care becomes a user of it.

The original recipient's entry also rises, by 8.4 points for help from children and 6.0 points for home services, but mostly when both partners start receiving the channel together. Joint entry accounts for 5.4 of the 8.4 points for help from children and 5.3 of the 6.0 points for home services (third column of Table 5). The remaining difference for the recipient alone is about 3 points for help from children and less than 1 point for home services. Most of the additional entry by the original recipient therefore occurs jointly with entry by the caregiver, and a relationship in which one person received care becomes a household in which two people newly receive help. Outside support is thus divided between two needs. Part of it addresses the caregiver's own new limitations rather than the care that the recipient previously received from the partner, which matters for whether that care has been replaced (Section 4.2).

The direct comparison between the two roles is less precise. The caregiver-minus-recipient differences are positive in every channel but not statistically significant, so the estimates do not rank the two partners' entry. Entry also understates the recipient's coverage, because baseline users cannot enter: at baseline, 17.1 percent of recipients already receive help from children and 32.2 percent receive home services, compared with 9.9 and 8.9 percent of caregivers. The evidence that new help is directed toward the caregiver is strongest for help

Table 5: Individual entry into support: any new caregiver limitation versus none

	Original caregiver	Original recipient	Both partners enter	Caregiver minus recipient
Help from children	13.35***	8.39**	5.36	4.97
	(4.15)	(4.24)	(3.39)	(4.59)
Entry rate, no new limitation (%)	13.33	15.22	7.03	
Observations			1,528	
Caregivers with any new limitation			112	
Home services	9.95***	5.95*	5.31**	4.01
	(3.10)	(3.35)	(2.33)	(3.43)
Entry rate, no new limitation (%)	8.26	12.12	3.66	
Home personal care	10.16***	4.27	2.04*	5.89
	(2.66)	(3.39)	(1.08)	(4.20)
Entry rate, no new limitation (%)	1.00	11.63	0.21	
Observations			1,890	
Caregivers with any new limitation			137	

Notes: Adjusted percentage-point differences in entry, with couple-bootstrap standard errors below estimates. Entry equals one when a person receives the channel at follow-up but not at baseline; baseline users have entry equal to zero. *Both partners enter* equals one when both original partners move from no receipt to receipt in the channel; it is contained in each partner's entry, so entry by the recipient alone equals the recipient column minus this column. *Entry rate, no new limitation* is the overlap-weighted percentage among observations whose caregiver develops no new limitation; adding the coefficient above it gives a model-implied entry rate with a new limitation, standardized to the weighted comparison group's baseline characteristics. Within each outcome, all columns use identical observations, weights, controls, and bootstrap draws, including both partners' baseline receipt. The final column directly estimates the role difference. There are 999 successful draws for help from children and home services, and 998 for home personal care. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

from children. Among couples in which neither partner initially receives this help, multiple new limitations raise the caregiver's entry by 20.7 points and the recipient's by 2.2 points, a difference of 18.5 points (s.e. 10.2; Appendix Table C.3).

A further concern is that joint or proxy answers may assign help to the wrong partner. When both partners answer their own social-support questions at both interviews, the caregiver's child-help entry difference is 14.8 points, close to the main estimate. Home personal-care entry is more sensitive to reporting: excluding the caregiver's proxy-only health-care answers reduces the estimate to 9.3 points when the restriction applies at follow-up and to 6.3 points when it applies at both interviews (Appendix Table C.5). The caregiver's new receipt of personal care therefore remains positive, but its size is less firmly established than that of new help from children.

4.2 More help, but not enough

When the caregiver develops multiple limitations, the original recipient more often reports that help falls short. Table 6 examines recipients who still have ADL limitations and live in the community, using the 1,381 observations with valid household-support and adequacy information. Reports that help is not consistently sufficient are 28.2 percentage points more common than in the comparison group, whose weighted rate is 26.9 percent: the share reporting a shortfall roughly doubles, from about one in four to more than one in two. Put differently, among 100 recipients with continuing limitations whose caregiver develops multiple limitations, about 28 more report that help is not enough than among comparable recipients whose caregiver stays free of limitations. With one new limitation, reported inadequacy does not increase (-4.0 points, s.e. 5.3), mirroring the concentration of partner-care exit among caregivers with multiple limitations.

This shortfall arises even though outside involvement has increased. The same households more often receive both outside channels (23.0 points, Panel A), and the joint occurrence of both-channel receipt and insufficient help is 16.0 points more common, rising from 5.4 percent in the comparison group to about 21 percent. Among 100 such recipients, about 16 more report that help is not enough while their household receives help from both children and home services.

The shortfall concerns help that is received but insufficient, not help that is absent. Panel B of Table 6 divides the inadequacy indicator into two exclusive parts. With multiple new limitations, reports of no activity help differ by only 4.6 points (s.e. 4.9), whereas reports that help is received but does not always meet needs are 23.6 points more common. Almost all of the inadequacy difference therefore arises among recipients who are receiving help. Together with the finding in Section 3 that additional partner-care exit does not measurably increase the share of recipients with no personal-care provider, this result locates the shortfall primarily among recipients who receive help, rather than among those receiving none.

Two explanations fit this pattern; the data locate the shortfall but cannot separate the hours, tasks, quality, or timing of the help received. First, part of the new outside support goes to the caregiver (Section 4.1), so an increase in household support overstates the help directed to the original recipient. Second, outside help differs from a partner's in availability. A partner is present throughout the day and can help whenever a need arises, with little cost of coordination. Children who live elsewhere and scheduled home services provide help at particular times. Outside help can therefore replace whether someone helps without replacing how much help is available or when it is available. Counting households as covered because they use both outside channels would miss the recipients in these households who

Table 6: Household support and recipient help adequacy on the same observations

	Rate, no new limitation (%)	Any new limitation	One new limitation	Multiple new limitations
A. Household support				
Both channels	17.73	12.36***	6.69	23.01***
		(4.57)	(5.31)	(8.01)
B. Recipient help adequacy				
Help not consistently sufficient	26.93	7.16	−4.02	28.18***
		(4.90)	(5.34)	(8.11)
No activity help	4.12	0.53	−1.63	4.58
		(2.31)	(2.48)	(4.90)
Received, not always sufficient	22.81	6.63	−2.39	23.60***
		(4.77)	(5.24)	(8.13)
C. Joint outcome				
Both channels and insufficient help	5.39	5.76*	0.33	15.97**
		(3.31)	(3.53)	(6.85)
Observations		1,381		
Caregivers with new limitations		103	63	40

Notes: Adjusted percentage-point differences relative to zero new limitations, with couple-bootstrap standard errors below estimates. The first column reports the overlap-weighted percentage among observations whose caregiver develops no new limitation; adding a coefficient gives a model-implied rate standardized to the weighted comparison group's baseline characteristics. Recipients are alive, interviewed in the community, still ADL-limited, and have valid adequacy and household-support answers. Every outcome uses identical observations, weights, and controls. Both models include baseline recipient help adequacy as adequate, inadequate, or unavailable. The two indented rows in Panel B partition the first row: their coefficients sum to the first row's coefficient up to rounding, and both use all 1,381 observations. The any-limitation model has 997 successful draws and the severity model 996, out of 999. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

report that help is not enough.

The result holds across samples and measures. In the broader sample of 1,423 observations with valid adequacy information, the difference adjusted for the recipient's baseline help adequacy is 29.1 points (s.e. 8.1). Under a stricter definition that counts only no help, or help that meets needs sometimes or hardly ever, as insufficient, the difference is 13.6 points (s.e. 6.6), nearly double the comparison-group rate of 7.2 percent. The reports come mainly from recipients themselves rather than from proxy respondents, who may include the caregiver: 29 of the 41 multiple-limitation observations are self-reported, and 18 of these 29 recipients (62 percent) report inadequate help, compared with 27 percent of self-reported recipients in the comparison group (253 of 929). Recipients who already received outside help also tend to keep it: among them, multiple caregiver limitations lower the probability of losing all measured support by 21.5 points (s.e. 8.0; 20 multiple-limitation observations; Appendix

Table D.3). Without the baseline adequacy adjustment, the multiple-limitations difference remains significant after Holm adjustment for multiple testing (adjusted $p = 0.0006$; Appendix Table D.1).

4.3 Gender and regional differences

Responses also differ by the caregiver's gender and by region. Table 7 reports partner-care exit, household receipt, and individual entry for multiple new limitations versus none, separately by group. Two differences stand out: partner-care exit rises much more when the caregiver is a man, and regions differ both in partner-care exit and in the caregiver's entry into home personal care.

Panel A of Table 7 shows that, with multiple new limitations, partner-care exit is 32.9 percentage points more common for male caregivers and 9.6 points more common for female caregivers. The male-minus-female difference of 23.3 points (s.e. 9.0) remains significant after adjustment within the family of care-transition outcomes (family-adjusted $q = 0.043$), though not across all interaction tests (global $q = 0.105$). Most couples are opposite-sex, so the comparison largely distinguishes husbands caring for wives from wives caring for husbands. Outside support does not offset this difference: household receipt and individual entry rise by similar amounts for male and female caregivers (Panels B and C). When a husband who cares for his wife develops multiple limitations, the existing arrangement therefore ends more often without a correspondingly larger outside response. The additional exit for male caregivers does not mainly reflect a handoff to another provider; much of it consists of exits after which no provider can be identified (Appendix Table E.1).

The clearest regional difference concerns home personal care for the original caregiver. With multiple new limitations, the caregiver's entry is 38.1 percentage points more common in Northern and Western Europe and 33.6 points more common in Southern Europe, compared with 1.9 points in Central and Eastern Europe. The regional interaction remains significant after adjustment for multiple comparisons (family-adjusted $q = 0.010$; global $q = 0.046$). The contrast between Northern and Western and Central and Eastern Europe does not reflect differences in initial service use or a single country: it persists after conditioning on the recipient's baseline personal-care receipt and when each country is excluded in turn (Appendix Table E.2). The Southern estimate, however, comes mostly from Spain, which contributes eight of the ten multiple-limitation observations. A broad home-service indicator would miss this distinction: households can obtain domestic assistance or meals without direct personal assistance reaching the person whose functional difficulties threaten the existing care arrangement.

Table 7: Partner-care exit and outside support by caregiver gender and region: multiple new limitations versus none

	Original caregiver		Region		
	Male	Female	North/West	South	Central/East
A. Original partner-care relationship					
Partner-care exit	32.89***	9.57	23.51***	37.13***	−2.18
	(6.12)	(6.96)	(7.83)	(7.47)	(8.46)
B. Household receipt at follow-up					
Help from children	26.58***	22.58***	33.16***	29.71***	7.31
	(8.82)	(7.11)	(8.79)	(10.84)	(9.52)
Home services	20.51**	21.10***	30.69***	8.49	18.43*
	(8.37)	(6.51)	(6.43)	(11.55)	(10.81)
Both channels	23.40**	24.08***	38.96***	15.46	10.57
	(9.25)	(6.65)	(8.86)	(11.78)	(9.05)
C. Individual entry into support					
Help from children					
Original caregiver	22.23	18.82**	28.42**	4.12	21.65*
	(15.45)	(8.02)	(12.51)	(12.66)	(11.91)
Original recipient	13.25	1.68	7.72	11.60	−1.38
	(14.49)	(8.23)	(12.47)	(17.34)	(10.77)
Home services					
Original caregiver	23.79**	16.69**	25.61***	21.31*	10.26
	(11.45)	(7.33)	(8.97)	(12.58)	(10.87)
Original recipient	19.61**	19.42**	18.59**	37.97**	11.33
	(9.74)	(8.39)	(8.68)	(15.56)	(9.86)
Home personal care					
Original caregiver	32.50***	19.27***	38.10***	33.64**	1.89
	(10.94)	(7.42)	(9.92)	(13.87)	(3.88)
Original recipient	−0.32	14.12*	9.19	19.42	4.11
	(7.17)	(7.89)	(9.02)	(15.90)	(7.83)

Notes: Adjusted percentage-point differences, with couple-bootstrap standard errors below estimates. Gender and regional estimates come from separate pooled interaction models using multiple-versus-zero overlap weights. Panel A measures partner-care exit; Panel B measures receipt by either partner; Panel C measures individual entry, with baseline users counted as nonentrants. Home personal care is a component of home services. Exit models use 2,445 observations, including 104 with multiple limitations: 49 male and 55 female caregivers, or 44 North/West, 32 South, and 28 Central/East. Household models use 2,406 observations, including 89 with multiple limitations: 37 male and 52 female caregivers, or 36 North/West, 26 South, and 27 Central/East. Child-help entry uses 1,460 observations with 44 multiple-limitations observations (14/30 by gender; 15/13/16 by region). Service and personal-care entry use 1,804 observations with 51 multiple-limitations observations (18/33; 22/10/19). Successful draws out of 399 are 397 for exit, 398 for household receipt and service entry, 387 for child-help entry, and 381 for personal-care entry. Stars test subgroup coefficients against zero, not differences between groups: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The male-minus-female exit contrast is 23.32 (s.e. 9.00), with family-adjusted $q = 0.043$ and global $q = 0.105$. The joint test of equal regional exit coefficients has $\chi^2(2) = 13.57$, family-adjusted $q = 0.038$, and global $q = 0.051$. The caregiver home-personal-care regional interaction has family-adjusted $q = 0.010$ and global $q = 0.046$. Appendix E reports the interaction tests and multiple-comparison adjustments.

In Central and Eastern Europe, additional help reaching the caregiver is concentrated in assistance from children rather than home personal care. Help from children to the caregiver rises by 21.7 points (s.e. 11.9), while home personal care barely changes; recipients in this region are also less likely to receive home personal care at baseline (17.5 percent, compared with 28.3 percent in Northern and Western Europe). Partner-care exit does not increase there (-2.2 points, s.e. 8.5), compared with increases of 23.5 points in Northern and Western Europe and 37.1 points in Southern Europe. The hypothesis that exit differences are equal across the three regions is rejected within the family of care-transition outcomes (family-adjusted $q = 0.038$), though not quite across all interaction tests (global $q = 0.051$). This pattern fits the larger role of family care in these care systems (Barczyk and Kredler, 2019; Suanet et al., 2012): where formal personal care is less available, a caregiver who develops multiple limitations more often continues to provide care, supported by children rather than by services. Household receipt and the other role-entry outcomes do not show clear regional differences after adjustment for multiple comparisons.

5 Further Evidence and Robustness

This section asks whether the changes persist, how they are timed relative to the recorded decline in caregiver health, and whether the main results depend on specification, reporting, sample selection, or the definition of health deterioration. Section 5.3 gives an overview of the robustness checks and their results; the appendix gives the full specifications and estimates.

5.1 Do the changes persist?

Table 8 follows partner care and the caregiver's receipt of home personal care to a second interview, about five years after baseline. Each comparison uses the same observations at both dates, keeping the original roles and first-follow-up health groups fixed.

The change in roles persists. For any new limitation, partner-care exit is 11.8 percentage points more common at the first interview and 12.8 points more common at the second. The caregiver's receipt of home personal care remains higher by 15.4 and 12.6 points. The broader home-service difference for the caregiver narrows, from 16.7 points at the first interview to 4.3 at the second (change -12.4 points, s.e. 5.9; Appendix Table F.1). What persists is hands-on personal care rather than broader service use. This persistence does not simply track the caregiver's health: in this personal-care sample, 31 of the 62 caregivers with a new limitation report no ADL limitation at the second interview. Once the caregiver has begun receiving help and partner care has ended, the new arrangement tends to remain in place.

Table 8: Partner-care exit and caregiver home personal care at two follow-ups

Outcome and health comparison	First follow-up	Second follow-up	Observations (new limitations)
Partner-care exit: any new limitation	11.8**	12.8**	988 (91)
	(5.6)	(5.2)	
Partner-care exit: multiple new limitations	9.2	12.9*	930 (33)
	(8.5)	(7.6)	
Caregiver home personal care: any new limitation	15.4***	12.6***	712 (62)
	(4.5)	(4.6)	

Notes: Adjusted percentage-point differences relative to no new caregiver limitations, with couple-bootstrap standard errors below estimates. Home personal care measures individual receipt at each interview, including receipt already present at baseline. Each row uses identical observations, overlap weights, and controls across interviews. Health groups are fixed at the first follow-up. Successful draws out of 999 are 998 for any-limitation exit, 873 for multiple-limitation exit, and 988 for personal-care receipt. The exit samples retain valid regular first-follow-up reports in wave 7. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Appendix F reports the full two-interview comparisons, health changes, and support transitions.

The multiple-limitations exit difference in this sample is smaller and less precise than in the main analysis: 9.2 points at the first interview, compared with 20.4 points in the full first-follow-up sample. The comparison requires both partners to remain observable through two follow-ups, and couples with multiple caregiver limitations often do not. Among the 171 such observations with a second scheduled interview, the original recipient has died in 81, compared with 1,210 of 3,469 observations without new limitations. Only 33 multiple-limitation observations remain, so the common sample has little power to detect a change in this difference, and the estimates give no clear evidence that it narrows or widens.

Counts of individual transitions point in the same direction (Appendix F). Partner care rarely resumes after it ends: among first exits with a known later care status, partner care resumes in 4 of 21 observations with multiple caregiver limitations, compared with 128 of 429 without new limitations. New help to the caregiver also tends to continue. Among caregivers with any new limitation and a known later community status, 14 of 15 who first received help from children and 11 of 13 who first received home services receive it again at the second interview. These are unweighted counts rather than adjusted comparisons.

5.2 Timing around the recorded decline

The timing comparison follows the same couples through three interviews. Baseline establishes the original relationship, caregivers report no ADL limitations at the earlier follow-up, and limitations reported at the later follow-up define the health comparison. This compari-

son shows how care changes before and after the caregiver's decline is recorded.

Partner care already weakens before the decline is recorded. At the earlier interview, when these caregivers still report no ADL limitations, partner-care exit is 11.3 percentage points more common than among caregivers who will remain free of limitations. The difference does not grow further once the limitation is recorded (3.2 points at the later interview; later-minus-earlier difference −8.0 points, s.e. 8.8). A broader comparison of all caregivers without current limitations gives the same pattern: exit is 9.7 points (s.e. 4.6) more common when a limitation is recorded at the next interview (Appendix Tables G.1 and G.2).

Outside support follows a different path. Before the decline is recorded, joint receipt of help from children and home services is, if anything, lower among these couples (−8.6 points, s.e. 3.9). Between the earlier and later interviews, joint receipt rises by 12.2 points relative to the comparison group (s.e. 6.9) and home-service receipt by 13.8 points (s.e. 7.8); these changes are imprecise, and their percentile intervals include zero. The interviews cannot date changes within each interval, but the point estimates describe a clear sequence: partner care erodes as caregiving capacity declines, and outside support increases once limitations are reported. The gap between the two is consistent with the reports of insufficient help in Section 4.2. Because the recorded ADL change marks a point in a longer process, the main estimates, which compare health and care at the same interview, describe the arrangement reached by then rather than the onset of the adjustment.

5.3 Robustness checks

This subsection reports additional checks that address four concerns: that the results depend on the model specification, on how survey questions were worded, on missing answers and survival, or on how caregiver decline is defined. I re-estimate the main comparisons with flexible machine-learning adjustment and logit outcome models, exclude the early waves that combine nursing and personal care in one question, use individually worded baseline questions, reweight for outcome nonresponse, examine selection through death and institutional residence, and replace the ADL count with a task-weighted difficulty index and with newly reported acute medical events. The exit gradient and the main support patterns remain under alternative adjustment methods and health definitions. The comparison between the two partners is sensitive to how baseline help is measured, and acute medical events reproduce the expansion of outside support, while their association with partner-care exit is imprecisely estimated. Table 9 reports the estimates, each with the corresponding estimate from the main analysis, and the paragraphs below discuss them.

Table 9: Additional robustness checks

Check and outcome	Health comparison	Adjusted difference	Reference estimate	Observations
A. Specification				
ML adjustment: household both-channel receipt	A	12.01***	11.14	2,537
		(3.18)		
ML adjustment: caregiver entry into help from children	A	13.43***	13.35	1,528
		(4.21)		
B. Questionnaire wording				
Baseline waves 4–8: household both-channel receipt	M	25.76***	23.79§	2,181
		(6.07)		
Individual wording: child-help entry, G minus R	M	−3.39	–	879
		(10.02)		
C. Alternative health definitions				
Task-weighted score of at least four: partner-care exit	W	19.85***	20.42	2,581
		(5.51)		
Acute medical events: partner-care exit	E	6.86	6.75	2,558
		(4.98)		
Acute medical events: household both-channel receipt	E	9.17**	11.14	2,506
		(4.09)		
Acute medical events: caregiver home-service entry	E	11.22**	9.95	1,860
		(4.70)		

Notes: Estimates and standard errors are percentage points. A compares any new caregiver ADL limitation with none; M compares multiple new limitations with none. W reports the higher weighted-score group relative to zero, jointly estimating the lower-score group and retaining any-limitation overlap weights. E compares newly reported acute medical episodes with no event. G and R denote the original caregiver and recipient. The reference estimate is the corresponding main estimate for the same outcome: the any-new-limitation estimate for A and E rows, and the multiple-limitations estimate for M and W rows. § The reference is the multiple-versus-zero benchmark with separately estimated weights on all baseline waves. Dashes indicate that no directly comparable main estimate is reported. Checks reported in Sections 3 and 4 and the nonresponse weighting are not repeated here; Appendix Tables B.3, B.5, C.5, D.1, and Appendix B.3 report them. Standard errors use couple-level bootstrap refitting, except ML estimates, which use grouped influence-score inference. Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Full specifications and inference details appear in the appendix.

Specification. The main estimates rely on a parametric propensity model and a linear outcome model. Flexible adjustment, using health-risk predictions estimated on the wider SHARE population and cross-fitted treatment and outcome models, gives household joint receipt of 12.0 points and caregiver child-help entry of 13.4 points, close to the main estimates (Panel A; Appendix Table H.3). Replacing the linear probability model with weighted logit outcome models changes the standardized probability differences little for exit, household support, and the caregiver-minus-recipient entry difference on identical observations (Appendix Table B.2). Too few bootstrap draws converge for these logit models, so they serve as point-estimate comparisons only.

Questionnaire wording and reporting. Early SHARE waves combine home nursing and personal care in one question. Restricting baseline waves to 4–8 leaves the multiple-limitations joint-receipt difference at 25.8 points (Panel B; Appendix Table B.4). The comparison between the two partners is the most sensitive result. When baseline child help is measured with individually worded questions, the multiple-limitations caregiver-minus-recipient entry difference is −3.4 points (s.e. 10.0; Appendix Table C.3). This supports the reading in Section 4.1: the evidence establishes new help to the caregiver more firmly than any ranking of the two partners' entry.

Outcome nonresponse. Some eligible couples lack valid answers to the outcome questions. Reweighting observed outcomes by the inverse of the estimated probability of a valid answer, which depends on caregiver health and baseline characteristics, changes the estimates for exit, joint receipt, and caregiver child-help entry by less than 0.1 percentage point (Appendix B.3). Outcome nonresponse is limited within these samples: the smallest fitted probability of a valid answer is 0.83.

Survival and residence. Household-support, entry, and adequacy results concern couples who remain together in the community. Caregiver decline makes this less likely. Among observations with multiple new limitations, 34.5 percent fail the community-couple conditions, compared with 25.9 percent without new limitations, mainly because the original recipient has died (55 of 197, compared with 798 of 4,245; Appendix D). If recipients with the greatest care needs are the most likely to die or enter institutions after caregiver decline, those who remain are relatively less dependent. The inadequacy difference in Section 4.2 would then be more likely to understate than overstate the shortfall among all original recipients. The data cannot determine what happened to the support of recipients who leave the sample: worst-case identification ranges for the loss of their baseline support include both large increases and large decreases. The paper therefore draws no conclusion about support loss outside intact community couples.

Alternative health definitions. Weighting each activity by its relative importance in the Barthel index preserves the severity pattern: at a threshold requiring more than one difficulty, the exit difference is 19.9 points (Panel C). Newly reported heart attacks, strokes, and hip fractures provide a comparison not defined by ADL reports. They are associated with greater joint receipt (9.2 points) and with the caregiver's entry into home services (11.2 points), although entry into help from children rises at least as much for the recipient as for the caregiver. The exit difference after an acute event is similar in size to that for any new

ADL limitation but imprecisely estimated (6.9 points, s.e. 5.0). Because a medical event need not impair the ability to provide care, the main analysis defines caregiver decline by ADL limitations (Appendix H).

6 Conclusion

Care at home often depends on a partner who is aging too. This paper shows what happens when that partner begins to need help. Following existing personal-care relationships in SHARE, I find that partner-care exit rises sharply only with multiple new limitations: one new caregiver limitation leaves partner care largely in place, while multiple new limitations make the original relationship about 20 percentage points more likely to end. Outside help responds as a combination of help from children and home services, and much of it goes to the former caregiver; most of the additional help to the original recipient starts together with the caregiver's. When the caregiver develops multiple limitations, original recipients with continuing ADL difficulties become far more likely to report that help does not consistently meet their needs, mostly because the help they receive is not enough. Part of the new support addresses the caregiver's own needs, and outside help provided at particular times may not reproduce the continuous availability of a co-resident partner. More outside support around a couple therefore does not mean that the original recipient's lost care has been replaced.

For the economics of family care, the central lesson is that an available partner is not a fixed stock of assistance. A healthy spouse provides informal insurance against long-term care needs, but the person who provides it is exposed to the very risk it covers, and the protection weakens sharply once the caregiver has multiple limitations. In Barczyk and Kredler (2018), for example, a disabled husband receives care from his healthy wife, and for tractability the husband is assumed to die once the wife becomes disabled, so the transition studied here does not arise. When it does arise, partner care often ends, new outside help is shared between the two partners, and the original recipient's needs are less often met. How much insurance a spouse provides depends on this transition. Estimates of whether formal care substitutes for family care face a related problem: when the caregiver also needs help, outside support can expand without replacing the care that was lost. Evaluating the protection that families provide, and the extent to which formal care substitutes for it, requires following the health, assistance, and help adequacy of both partners.

The evidence points to three lessons for long-term care policy. First, a decline in the caregiver's health should prompt a reassessment of both partners, not only of the person who first applied for support. The assessment needs to establish which tasks the caregiver can still provide, what assistance the caregiver now needs, and whether the original recipient

still receives enough help. Some systems already move in this direction: the French needs assessment for the Allocation personnalisée d'autonomie considers the support needs of close caregivers.[4] The timing of reassessment also matters. The point estimates suggest that partner care weakens before the caregiver's limitation is recorded, while outside support rises afterwards, so a reassessment that waits until the caregiver's own limitations are formally established may come too late. Arrangements in which a husband cares for his wife deserve particular attention, because they end far more often after the caregiver's decline without a larger outside response. Second, the success of a care plan should be judged by whether each person's needs are met, not by whether the household uses services. In the households studied here, greater involvement from both children and home services accompanies more frequent reports of insufficient help, mostly among recipients who already receive help. How much help is available, and when, matters as much as whether any help is received. Third, reassessment must take account of the local mix of family help and services. In Central and Eastern Europe, caregivers who develop multiple limitations rarely start receiving home personal care; instead, partner care more often continues, supported by help from children. The assistance available to each partner, and the burden that remains on the declining caregiver, cannot be assumed to be the same across care systems.

The findings also show what data are needed to monitor care at home. Identifying who receives new help required following both partners over time and assigning receipt to a specific person. In early SHARE waves, questions about help from outside the household refer jointly to the respondent and partner, so a positive answer identifies household receipt but not its destination. Useful data would record who helps whom, with which tasks, for how many hours, and whether each person's needs are met. Questions about sufficiency should be asked of people who already receive help, since the shortfall documented here mainly concerns help that is received but insufficient.

Richer data would allow these results to be taken further. Information on hours and tasks would show how much of the lost partner care outside help replaces, and whether the shortfall reflects too few hours of help or help that is not available when it is needed. Linking survey reports to administrative care assessments would show whether, and how quickly, care systems respond when the caregiver's health declines. Variation in the availability of home care, such as the expansion of long-term care entitlements under Spain's 2006 dependency law,[5] could show whether services that reach the declining caregiver also restore adequate help for the original recipient. Following couples beyond the community, into institutions

[4]See the official CNSA guidance, "Comment les besoins sont-ils évalués?". The assessment includes the older person's needs, existing assistance, living environment, and the support needs of close caregivers.

[5]Ley 39/2006, de 14 de diciembre, de Promoción de la Autonomía Personal y Atención a las personas en situación de dependencia. See Costa-i-Font et al. (2025) for evidence on the reform.

and through bereavement, would complete the picture of how care is reorganized when the person who provides it begins to need it. As populations age, more care at home will depend on partners who are themselves growing old. Supporting these couples requires recognizing that the person who supplies care can soon come to need it.

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Appendix

The appendix follows the empirical argument: measurement and sample construction; estimation and baseline adjustment; support destinations and reporting; recipient adequacy and follow-up states; gender and regional comparisons; later care and support; timing; and alternative health measures. Tables identify whether the exposure is any new limitation or multiple new limitations. Original caregiver and recipient roles remain fixed throughout.

A Measurement and Sample Construction

A.1 Questionnaire items and harmonization

Table A.1: From original survey items to the support comparisons

Concept	Source items	Construction and analysis variable
Baseline care direction	SP018–SP019 provision; SP020–SP021 receipt	Personal care reported from the zero-ADL partner to the partner with positive ADL. One positive report establishes the relation; `care_both` records confirmation by both.
Caregiver transition	Six ADL items in PH049	Baseline count is zero. Follow-up count is at least one for $D = 1$ and zero for $D = 0$. The severity comparison separates one from multiple new limitations.
Help from children	SP002 receipt and SP003 outside-helper relationships; SP020–SP021 household helpers	Help from children or stepchildren living elsewhere, or personal care from them within the household. `hh_child` records receipt by either original partner.
Home services	HC032 in waves 1–2; HC127 from wave 5	Selected categories 1, 2, and 3 identify home personal care, domestic assistance, and meals. `hh_services` records any of these for either partner.
Household configuration	The two household indicators	`hh_both`, `hh_child_only`, `hh_service_only`, and `hh_neither` are the four combinations of valid child and service states.
Role-specific entry	Valid baseline and follow-up receipt for each person	`g_new_child` and `r_new_child` are zero-to-one transitions for the fixed roles. Service entry uses the same rule. `d_new_child` is caregiver minus recipient entry.

Notes: Item names refer to the original SHARE questionnaires. Helper relationship codes are wave-specific. Help from children includes broad help from children living elsewhere and household personal care; home services are receipt measures, not payer measures. A positive establishes receipt; a zero requires sufficient valid component information or documented routing. Wave 4 has no comparable home-service item. At wave 7, items not asked in the life-history questionnaire remain missing. Household follow-up waves are 2, 5, 6, 7, 8, and 9; role entry additionally requires identifiable individual states at both interviews. An early joint positive is not assigned to either person.

The panel is built from the original physical-health, social-support, health-care, cover-screen, and end-of-life modules. Availability is determined by valid responses and docu-

mented routing, not by module linkage. Negative missing codes, refusals, unasked questions, and unresolved provider identities remain distinct from valid zero responses. A definite questionnaire skip after a valid health or household gate can establish no receipt; an unresolved gate or missing response cannot.

PH049 supplies the six ADL items and the common seven IADL items, which exclude items 14 and 15 added from wave 6. Item counts require valid information on the full scale; published generated counts are retained only for comparison. SP018 and SP019 identify personal care given within the household and SP020 and SP021 care received, with relationship codes mapped by wave. In wave 4, named social-network members are resolved through the network relationship information rather than treated as ordinary relationship categories.

Help from children living elsewhere combines the SP002 receipt gate with SP003 helper relationships and is not restricted to personal-care tasks; the within-household component uses SP021 child or stepchild codes. A positive from either component establishes receipt, and a negative requires both components to be observed as negative. Home services use HC032 in waves 1 and 2 and HC127 from wave 5. The household service indicator is negative only when the component information establishes no selected service for either partner. The wave-2 outside-help questions can refer to the period since the previous interview, whereas later regular questions generally refer to the previous twelve months.

Under the early joint wording of outside-help questions, a household positive is retained without assigning it to either partner. The paired child-entry sample therefore excludes unresolved baseline role states and jointly worded follow-up states. A joint baseline negative is used only when household evidence establishes no help from children living elsewhere and there is no contrary positive.

Both entry roles are observed on a channel-specific common sample. No control is added for the recipient's follow-up health state, partner-care exit, or later provider arrangement in the main household or entry models, because those variables can change during the interval and are not baseline confounders. The indicators describe the original couple's support even when the recipient no longer reports an ADL limitation. These binary measures identify who receives support, not the quantity of partner care replaced.

A.2 From the respondent panel to baseline care observations

Table A.2 records the construction of the baseline care sample. Counts before follow-up scheduling are couple-wave records and can include repeated observations for a couple. The resulting 6,359 observations begin with an actual directional personal-care relationship.

At scheduled follow-up, caregiver health is classified in 4,722 observations: 4,245 with

Table A.2: Construction of the baseline care sample

Construction step	Records remaining
Person-wave panel	731,726
Candidate couple-wave links	214,109
Reciprocal links, common household and country, completed main interviews	142,839
Valid health information for both partners	141,564
One partner has positive ADL and the other zero	20,306
Personal care confirmed by provision or receipt	8,293
European records, excluding Israel	8,018
Exclude wave 9 baselines (1,365 records)	6,653
Exclude earlier jointly ADL-positive episodes (279 records)	6,374
Exclude records without a later country wave (15 records)	6,359

Notes: The first row counts person-wave observations; subsequent rows count candidate or retained couple-wave records. Follow-up is scheduled by country participation. A couple may contribute more than one baseline observation.

zero limitations, 280 with one, and 197 with multiple limitations. The remaining 1,637 comprise 246 caregiver deaths, 1,375 without a completed main interview, and 16 with missing ADL information. Outcome-specific samples then apply the response and attribution rules described in Section 2.2.

B Estimation and Outcome Nonresponse

The main propensity model has 11 nonconstant slopes. The paired entry models add two baseline-support indicators and have 13. Point estimates retain the specified sample without trimming. Effective sample size is $(\sum_i w_i)^2 / \sum_i w_i^2$, calculated separately by observed exposure group. Table B.1 records these quantities and bootstrap success.

Table B.1: Estimation-sample diagnostics

Sample	Control ESS	New-limitation ESS	Maximum country SMD	Successful draws
Household common sample	1,566	216	0.233	999
Child entry	1,012	110.5	0.282	999
Home-service entry	1,104	133.9	0.215	999
Personal-care entry	1,071	132.8	0.194	998

Notes: ESS denotes effective sample size. Maximum country SMD is the largest absolute standardized mean difference across country indicators, which are not included in the propensity model. Successful draws are reported out of 999 couple-level bootstrap resamples.

For the paired role analysis, the caregiver-minus-recipient contrast is estimated by re-

gressing the difference in entry indicators on the same observations, weights, and regressors as the two role-specific models. In the linear model, its coefficient equals the caregiver coefficient minus the recipient coefficient. The couple-level bootstrap re-estimates this difference regression in each draw. The four exclusive entry-state coefficients sum to zero in each draw.

The interaction models and multiple-comparison adjustments for gender and region are described in Appendix E.

B.1 Weighted logit functional-form comparisons

Table B.2 separates a change in the outcome response function from a change in the estimation sample. The propensity model, overlap weights, baseline controls, and country indicators follow the corresponding linear specification. Logit probability differences average predictions under each health state over the retained observations using the original-sample overlap weights. Interaction models average within each observed gender or region. For paired entry, the caregiver and recipient logits are fitted on their common retained sample, and the two standardized probability changes are subtracted.

On the retained samples, changing the response function has little effect on these estimates. Sample retention matters more for paired entry: ordinary logit drops perfectly predicted country cells and baseline users whose entry indicator is zero by definition, which explains most of the gap between the original-sample and common-sample paired estimates. The caregiver home-personal-care logits do not converge, so this outcome has no paired logit comparison. Couple-level bootstrap audits of the logit models fall well below the 80 percent success rule; for example, 443 of 999 draws succeed for any-limitation exit and 24 of 399 for paired child-help entry. The table therefore reports no logit standard errors or significance tests.

B.2 Separate severity weights and baseline-support adjustment

The multiple-versus-zero comparison refits its own propensity after excluding one-limitation observations (Table B.3). Tables B.4 and B.5 examine the early questionnaire waves and baseline household configurations. The latter holds the 1,760 observations fixed across adjustment sets.

B.3 Nonterminal outcome nonresponse

This sensitivity analysis weights for missing outcome answers among confirmed community survivors with an observed caregiver ADL classification. The exit sample permits the orig-

Table B.2: Linear and logit outcome models: point-estimate comparisons

Outcome and comparison	Original LPM	Common LPM	Common logit	Original N	Common N
A. Main outcomes and paired entry					
Partner-care exit (A)	6.75	6.75	6.58	2,581	2,581
Partner-care exit (M)	20.42	20.42	20.06	2,581	2,581
Household both-channel receipt (A)	11.14	11.27	11.24	2,537	2,474
Caregiver child-help entry (A)	13.35	14.94	14.93	1,528	1,342
Recipient inadequacy, baseline adjusted (M)	29.08	29.31	28.80	1,423	1,395
Child-help entry: caregiver minus recipient (A)	4.97	2.16	1.67	1,528	1,163
Home-service entry: caregiver minus recipient (A)	4.01	0.02	0.04	1,890	1,152
B. Selected gender and regional comparisons (M)					
Partner-care exit: male caregiver	32.89	32.89	33.14	2,445	2,445
Partner-care exit: female caregiver	9.57	9.57	9.30	2,445	2,445
Both-channel receipt: North/West	38.96	38.99	38.80	2,406	2,341
Both-channel receipt: South	15.46	15.43	15.47	2,406	2,341
Both-channel receipt: Central/East	10.57	10.87	11.45	2,406	2,341

Notes: Values are probability differences in percentage points. LPM denotes linear probability model; A compares any new limitation with none, and M compares multiple new limitations with none. Common-sample columns use identical observations, weights, and controls within each row. Paired-entry rows use the intersection of both roles' logit estimation samples. Interaction sample sizes refer to the pooled model, not the subgroup alone. Panel A severity estimates retain any-limitation overlap weights; Panel B uses separately estimated multiple-versus-zero weights. Logit entries are standardized discrete probability changes, not log-odds coefficients. The table reports point comparisons only; see the text for bootstrap convergence.

inal partners to live apart; household and role-entry samples require the original couple to remain linked and co-resident. Eligibility respects questionnaire availability: the exit analysis excludes follow-up wave 7, service outcomes exclude wave 4, wave-7 household outcomes require both regular questionnaires, and individual child entry requires individually worded follow-up questions. Entry models also require valid baseline receipt. Deaths, institutional residence, unknown survival or residence, and unavailable caregiver ADL remain separate states.

Let R_i indicate that the required outcome answers are valid within this eligible sample. Following inverse-probability-of-censoring weighting (Robins et al., 1995), estimate $q_i = \Pr(R_i = 1 \mid D_i, X_i)$ and use $w_i^{\text{IPCW}} = w_i^{\text{overlap}}/\widehat{q}_i$ for observed outcomes. The overlap propensity is fitted on the full eligible sample. The response model includes the any-new-limitation indicator, baseline caregiver age and seven-item IADL, recipient ADL, years between survey waves, and confirmation of baseline care by both partners. Because miss-

Table B.3: Primary exposure and the multiple-versus-zero binary benchmark

	Any new limitation	Multiple new limitations
Partner-care exit	6.75**	20.53***
	(3.25)	(4.77)
Household receipt: children	14.67***	24.25***
	(3.57)	(5.33)
Household receipt: services	9.46***	20.85***
	(3.31)	(5.40)
Household receipt: both channels	11.14***	23.79***
	(2.93)	(5.45)
G: entry into help from children	13.35***	19.87***
	(4.15)	(7.12)
G: entry into home services	9.95***	19.15***
	(3.10)	(5.51)
G: entry into home personal care	10.16***	23.82***
	(2.66)	(5.90)

Notes: Coefficients are percentage-point differences. Couple-bootstrap standard errors are below estimates. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Observations (new-limitation observations) for the any-new-limitation and multiple-new-limitations columns are: exit, 2,581 (240) and 2,445 (104); household receipt, 2,537 (220) and 2,406 (89); child-help entry, 1,528 (112) and 1,460 (44); home-service and home personal-care entry, 1,890 (137) and 1,804 (51). Each draw re-estimates both model stages; in the any-new-limitation column, exit, household receipt, and child-help and home-service entry have 999 successful draws; home personal-care entry has 998. In the multiple-new-limitations column, exit has 996, household receipt 999, child-help entry 978, home-service entry 998, and home personal-care entry 952 successful draws. The multiple-versus-zero column excludes one-new-limitation observations and refits its own binary propensity. It has a different overlap target from the primary exposure and from the common-weight severity decomposition. Differences between these columns are not formal severity contrasts.

Table B.4: Household receipt with baseline waves 4–8

	Help from children	Home services	Both channels
Baseline waves 4–8	24.42***	23.01***	25.76***
	(5.91)	(5.63)	(6.07)
Observations		2,181	
New-limitation observations		83	

Notes: All estimates are in percentage points. Bootstrap standard errors are in parentheses on the following line. Stars use two-sided normal-approximation tests based on those standard errors: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The sample compares multiple new limitations with zero and restricts baseline waves to 4–8. Both the propensity and outcome models are fitted on this sample, with 399 successful bootstrap draws for each outcome.

Table B.5: Household receipt conditional on baseline outside support

	Original adjustment	Baseline support added
A. Household receipt by channel		
Help from children	19.29***	16.95**
	(7.39)	(7.34)
Rate, no new limitation (%)	37.73	39.48
Home services	25.89***	25.06***
	(6.78)	(6.86)
Rate, no new limitation (%)	36.48	36.55
B. Household configuration (mutually exclusive)		
Both channels	22.13***	19.49***
	(6.97)	(6.60)
Rate, no new limitation (%)	17.00	18.50
Children only	−2.84	−2.54
	(5.71)	(5.67)
Services only	3.76	5.57
	(6.50)	(6.91)
Neither channel	−23.05***	−22.52***
	(5.78)	(5.85)
Rate, no new limitation (%)	42.79	42.47
Observations	1,760	1,760
New-limitation observations	50	50

Notes: Estimates are percentage-point contrasts for multiple new limitations versus none. Couple-bootstrap standard errors are below coefficients. Stars use two-sided normal-approximation tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Both columns use exactly the same observations with valid baseline household child and service states. The second column adds indicators for the four baseline configurations to both the propensity and outcome models; overlap weights are refitted. Rate rows report the overlap-weighted percentage among observations whose caregiver develops no new limitation under each specification. The two channels in Panel A can overlap; the configurations in Panel B are mutually exclusive, and their coefficients sum to zero up to rounding. There are 394 successful draws out of 399 in each model.

ing answers are sparse, a ridge logit uses six standardized slopes with penalty one and an unpenalized intercept. The penalty is fixed across outcomes and bootstrap draws. Outcome adjustment follows the main specification. The weighting assumes that response is independent of the missing outcome conditional on these predictors and that observation probabilities are positive.

Table B.6 separates the complete-case main estimator, overlap weights estimated on the eligible sample without response correction, and the combined weights. The complete-case coefficients reproduce the main tables. Each of 999 couple-bootstrap draws re-estimates both propensities, the response model, and the outcome regressions.

Table B.6: Nonterminal IPCW: partner-care exit, household receipt, and individual entry

	Complete-case overlap	Eligible-sample overlap	Eligible overlap plus IPCW	Observed / eligible
Partner-care exit: any new limitation	6.75**	6.73**	6.74**	2,581 / 2,605
	(3.20)	(3.20)	(3.20)	
Partner-care exit: multiple new limitations	20.42***	20.40***	20.42***	
	(4.55)	(4.55)	(4.55)	
Household receipt: help from children	14.67***	14.68***	14.66***	2,537 / 2,583
	(3.60)	(3.60)	(3.60)	
Household receipt: home services	9.46***	9.43***	9.43***	
	(3.49)	(3.49)	(3.50)	
Household receipt: both channels	11.14***	11.11***	11.08***	
	(3.15)	(3.14)	(3.14)	
Caregiver entry: help from children	13.35***	13.37***	13.31***	1,528 / 1,558
	(4.09)	(4.09)	(4.09)	
Caregiver entry: home services	9.95***	9.95***	9.95***	1,890 / 1,900
	(3.13)	(3.13)	(3.13)	
Caregiver entry: home personal care	10.16***	10.16***	10.15***	1,890 / 1,900
	(2.55)	(2.55)	(2.55)	

Notes: Coefficients are percentage-point differences; couple-bootstrap standard errors appear below. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Counts refer to observations; one couple can contribute more than one. New-limitation observations (observed / eligible) are exit: 240 / 242; household: 220 / 223; child: 112 / 115; service: 137 / 137; personal: 137 / 137. The multiple-limitations exit coefficient retains any-new-limitation overlap weights and is estimated jointly with one new limitation. Successful draws out of 999 are exit: 999; household: 999; child: 999; service: 998; personal: 997. The combined weights use fitted probabilities without trimming.

Outcome nonresponse is limited within these samples, as the observed and eligible counts in Table B.6 show. The smallest fitted observation probability is 0.83. In the household sample, the combined-weight effective sample sizes are 1,575 controls and 216 treated observations. The response correction leaves the main patterns essentially unchanged.

C Support Channels, Starting Receipt, and Reporting

C.1 Home services by task

Table C.1 separates personal care, domestic assistance, and meals. Task receipt can change within an existing service arrangement, so its contrast can exceed the contrast for any home-service receipt.

Table C.1: Household receipt of home services by task: any new limitation

	Rate, no new limitation (%)	Adjusted difference
Home personal care	24.61	10.57***
		(3.26)
Domestic services	28.98	8.65**
		(3.37)
Delivered meals	8.08	4.44*
		(2.45)
Observations	2,533	
New-limitation observations	218	

Notes: Coefficients are percentage-point differences. Couple-bootstrap standard errors are below estimates. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The first column reports the overlap-weighted percentage among observations whose caregiver develops no new limitation. Each draw re-estimates both model stages; all estimates have 999 successful draws. All task components must be observed. They overlap and are not additive.

C.2 Entry destinations and starting receipt

For each channel, caregiver entry equals caregiver-only entry plus entry by both partners; recipient entry equals recipient-only entry plus entry by both. Table 5 reports joint entry for any new limitation. Table C.2 provides the four mutually exclusive patterns for the separately weighted multiple-versus-zero comparison.

Baseline receipt differs between the roles: 9.9 percent of caregivers and 17.1 percent of recipients receive help from children, while 8.9 and 32.2 percent receive home services. Table C.6 restricts the any-new-limitation comparison to couples initially without the channel. Table C.3 separately reports multiple-limitation contrasts using a pooled interaction with neither partner initially receiving support. In that model, the propensity includes both baseline receipt indicators, and the outcome regression adds the both-zero indicator and its interaction with exposure. Summing the exposure and interaction coefficients gives the both-zero contrast. Child-help and service comparisons include 1,164 and 1,192 both-zero observations, respectively, within full samples of 1,460 and 1,804.

Individually worded baseline child-help questions restrict the multiple-limitations comparison to waves 6–8, with 879 observations and 24 caregivers with multiple limitations. The caregiver-minus-recipient contrast is −3.4 points (s.e. 10.0). Table C.4 instead requires both channels on the same observations and directly estimates the difference between their role contrasts.

Table C.2: Mutually exclusive patterns of support entry (multiple new limitations)

	Help from children	Home services
Caregiver only	13.47**	7.45*
	(5.81)	(4.01)
Rate, no new limitation (%)	7.53	4.50
Recipient only	−1.16	7.78*
	(4.27)	(4.59)
Rate, no new limitation (%)	7.91	6.67
Both partners	6.40	11.70**
	(5.74)	(4.67)
Rate, no new limitation (%)	7.62	3.87
Neither partner enters	−18.71***	−26.93***
	(7.18)	(6.38)
Rate, no new limitation (%)	76.93	84.96
Observations	1,460	1,804
New-limitation observations	44	51

Notes: Estimates are percentage points, with bootstrap standard errors below coefficients. Stars use two-sided normal-approximation tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The four outcomes partition each channel sample and their coefficients sum to zero up to rounding. Neither partner enters can include continued baseline support. Rate rows report the overlap-weighted percentage among observations whose caregiver develops no new limitation. The channel samples differ; cross-channel contrasts use Table C.4.

C.3 Section-specific answering status

Answering status is joined by person and wave using SP022 for social support, PH054 for physical health, and HC063 for health care. Codes distinguish respondent-only, mixed respondent-and-proxy, and proxy-only answers. Exit restrictions follow the baseline-selected reporter; a health-gate zero also requires that person's physical-health section to meet the restriction. Paired child-entry checks require both partners' social-support sections at both dates. Paired service checks use both partners' health-care sections, whereas the caregiver's own personal-care check restricts only that person's section.

HC063 is available in waves 1, 2, 8, and 9; other waves have unknown section status. Tables C.5 and C.6 report known-status reference samples alongside the restrictions. Prior caregiver ADL history enters the health-care outcome regression but is omitted from its propensity model to avoid separation. Inference requires the point fit to retain all specified observations and at least 80 percent of bootstrap draws to succeed.

The personal-care case review covers 19 observations involving 19 caregivers in 18 couples. All have a valid baseline zero and follow-up one on the original item. Six health-care

Table C.3: Support entry and changes in receipt under alternative baseline measures

	Help from children	Home services	Home personal care
A. Multiple new limitations: neither partner initially receives the channel			
Caregiver entry	20.71**	18.75**	17.44***
	(9.00)	(7.79)	(6.37)
Recipient entry	2.24	26.38***	11.88*
	(9.05)	(8.75)	(7.06)
Caregiver minus recipient	18.46*	−7.64	5.56
	(10.16)	(6.21)	(8.81)
Observations with neither partner initially receiving support	1,164	1,192	1,383
Of which: multiple new limitations	31	29	36
B. Caregiver-minus-recipient contrasts under alternative definitions			
Change in receipt: multiple new limitations	13.16	−5.82	13.83
	(8.07)	(6.18)	(8.58)
Entry: any new limitation	4.97	4.01	5.89
	(4.60)	(3.50)	(3.97)
Entry: multiple new limitations, individually worded baseline questions	−3.39		
	(10.02)		

Notes: Estimates are percentage points, with couple-bootstrap standard errors below coefficients. Stars use two-sided normal-approximation tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The upper panel uses the pooled both-zero interaction described in the text, not an isolated-subgroup propensity fit. Its full estimation samples contain 1,460, 1,804 and 1,804 observations. Change contrasts use these same channel samples and compare multiple new limitations with zero. Any-limitation entry uses 1,528 observations and 112 treated observations for children, and 1,890 and 137 for services. The individual-wording child sample contains 879 observations and 24 observations with multiple new limitations. Successful draws are 396, 398 and 380 for both-zero and change models; 399 for any new limitation; and 348 for individual wording, out of 399 requested in each case. Inference requires the bootstrap success threshold specified in Section 2.4.

sections are respondent-only, three mixed, two proxy-only, and eight lack the section-status field. Overall interview indicators flag proxy participation in 17 cases, including four with respondent-only health-care sections. Section-specific status therefore distinguishes answers to the outcome questions from proxy participation elsewhere in the interview.

Table C.4: Support entry by original role with both channels measured

	Caregiver	Recipient	Difference
Child entry	20.64***	6.92	13.72*
	(7.75)	(7.22)	(7.98)
Rate, no new limitation (%)	15.00	15.80	
Service entry	18.56***	20.96***	−2.40
	(6.64)	(6.34)	(7.51)
Rate, no new limitation (%)	7.58	9.51	
Child minus service role contrast			16.11*
			(8.64)
Observations		1,411	
New-limitation observations		38	

Notes: Percentage-point contrasts for multiple new limitations versus none, with couple-bootstrap standard errors below estimates. Stars use two-sided normal-approximation tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All rows use the same propensity and outcome covariates, including all four role-specific baseline support states. The last coefficient is estimated directly on $(N_G^{child} - N_R^{child}) - (N_G^{service} - N_R^{service})$, preserving within-couple and cross-channel covariance. Rate rows report the overlap-weighted entry percentage among observations whose caregiver develops no new limitation. There are 390 successful draws out of 399.

Table C.5: Partner-care exit and support entry by section answering status

	Status known	No proxy-only answers	Respondent only
Partner-care exit	6.81**	7.38**	6.10
	(3.38)	(3.68)	(3.89)
Observations	2,580	2,252	2,039
New-limitation observations	240	196	171
G: entry into help from children	13.35***	16.60***	14.84***
	(4.01)	(4.74)	(5.09)
R: entry into help from children	8.39**	9.90*	6.10
	(4.13)	(5.28)	(5.43)
Caregiver minus recipient	4.97	6.70	8.74
	(4.53)	(5.10)	(5.55)
Observations	1,528	1,235	1,061
New-limitation observations	112	77	63
G: entry into home services	12.27**	9.75*	8.73
	(4.99)	(5.88)	(5.71)
R: entry into home services	7.76	3.97	5.23
	(4.99)	(5.60)	(5.71)
Observations	787	637	512
New-limitation observations	54	39	32
G: entry into home personal care: own health-care section			
Follow-up restriction	10.99***	9.35***	5.92
	(3.65)	(3.55)	–
Observations	937	925	904
New-limitation observations	67	62	58
Successful draws	854	856	658
Both-date restriction	9.40***	6.32*	4.36
	(3.61)	(3.34)	–
Observations	787	772	743
New-limitation observations	54	47	43
Successful draws	877	852	616

Notes: Coefficients are percentage-point differences. Couple-bootstrap standard errors are below estimates. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. No proxy-only answers retains joint answers; respondent only excludes joint answers as well. The reporting restrictions and health-care propensity adjustment are described above. Each draw re-estimates both stages, with 999 draws requested and at least 800 successful draws required for inference. Exit has 999 successful draws in every column; child entry has 998, 976, and 950; home-service entry has 999, 983, and 927. Dashes indicate unavailable inference, not zero standard errors. Personal-care points are retained for the strictest restrictions without stars or standard errors because too few draws meet the inference threshold. Normal-test stars and percentile intervals can differ near significance thresholds.

Table C.6: Sensitivity of caregiver-only entry to starting receipt and reporting

Restriction	Caregiver-only entry	Rate, no new limitation (%)	Observations (new limitations)	Successful draws
Help from children				
Neither partner initially receives the channel	5.43*	5.17	1,218 (85)	952/999
	(3.23)			
Individually worded baseline questions	3.70	7.77	916 (61)	974/999
	(3.88)			
No full proxy, both interviews	8.59**	6.14	1,235 (77)	980/999
	(3.75)			
Respondent-only, both interviews	9.52**	6.05	1,061 (63)	939/999
	(3.99)			
Home services				
Neither partner initially receives the channel	−1.70**	1.95	1,249 (86)	942/999
	(0.79)			
Reporting status known, both interviews	3.76†	5.83	787 (54)	638/999
	–			
Reporting status known at follow-up	6.30	6.89	937 (67)	847/999
	(3.89)			
No full proxy at follow-up	6.60	6.19	788 (53)	873/999
	(4.15)			
Home personal care				
Full paired-entry sample	8.12***	0.80	1,890 (137)	998/999
	(2.23)			
Neither partner initially receives the channel	1.99	0.44	1,450 (103)	987/999
	(1.38)			
Reporting status known, both interviews	5.53†	1.34	787 (54)	537/999
	–			
Reporting status known at follow-up	8.02†	1.19	937 (67)	728/999
	–			
No full proxy at follow-up	7.62†	1.06	788 (53)	769/999
	–			

Notes: All comparisons use any new limitation versus none and retain the main propensity and outcome specifications. Coefficients and standard errors are percentage points; stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Caregiver-only entry means that G enters while R does not enter; R may continue receiving baseline support. The rate column reports the overlap-weighted percentage among observations whose caregiver develops no new limitation. Reporting restrictions use both partners' relevant social-support sections for help from children and health-care sections for home services. No-full-proxy restrictions retain mixed answers. Health-care answering status is unavailable in released waves 4–7. For home services and home personal care, restrictions requiring no full proxy or respondent-only answers at both interviews are omitted because their point propensity fits exclude observations. † marks a point estimate with fewer than 800 successful bootstrap draws, for which no standard error is reported. Each attempted bootstrap requests 999 couple-level draws and refits both models.

D Recipient Help Adequacy, Endpoints, and Existing Support

The endpoint classification distinguishes recipient death, institutional residence, ADL recovery, continued partner care, nonpartner personal-care handoff, no measured provider with continued ADL need, and unresolved status. Death and institution take precedence over community care. The zero-versus-multiple classification contains 4,442 observations, including 197 with multiple new caregiver limitations; its observable-exit sample contains 2,445 and 104, respectively. The baseline reporting person remains fixed, with the recipient used whenever that person confirmed baseline care. On the 2,581 observations in the any-new-limitation exit comparison, the adjusted exit difference is 6.75 percentage points (s.e. 3.25), while exit accompanied by recipient ADL recovery differs by 0.10 points (s.e. 3.00). Both use the main overlap weights, with 999 successful couple-bootstrap draws.

D.1 Recipient help adequacy

Table D.1 reports the broader recipient-adequacy comparison, retaining one and multiple new limitations in the same sample and using the main any-limitation overlap weights. The outcome is one when PH050 records no help with difficult activities, or PH051 records help meeting needs usually, sometimes, or hardly ever rather than all the time. The stricter measure combines no help with the latter two PH051 categories. Baseline adequacy, when included, uses the same questions and distinguishes adequate, inadequate, and unavailable answers. It is available for 870 of the 1,423 observations, so adjustment retains a separate category for the remaining baseline answers.

Absence of a personal-care provider is a separate endpoint. Restricting the matched-reporter exit sample to follow-ups in waves 6, 8, and 9, when all personal-care sources are structurally available, leaves 1,561 observations: 1,440 with zero new limitations, 73 with one, and 48 with multiple. The multiple-limitations difference in continued ADL need with no measured personal-care provider after partner-care exit is 1.0 percentage point (s.e. 4.9), based on 998 successful bootstrap draws. The corresponding any-limitation estimate is -2.1 points (s.e. 2.9). This distinguishes complete provider absence from the adequacy of help received.

The physical-health module also identifies who answered these questions. In the adequacy sample, 29 of the 41 multiple-limitations observations are self-reported, five mix respondent and proxy answers, and seven are fully proxy-answered. Eighteen of the 29 self-reported records indicate inadequacy, compared with 253 of 929 self-reported controls. These descrip-

Table D.1: Recipient help adequacy and adjustment for baseline help

	Any new limitation	One new limitation	Multiple new limitations
Help not consistently sufficient: main controls	8.19*	−3.78	30.32***
	(4.65)	(5.55)	(7.80)
Rate, no new limitation (%)		25.08	
Help not consistently sufficient: baseline help added	7.83	−3.75	29.08***
	(4.82)	(5.50)	(8.11)
Rate, no new limitation (%)		26.69	
No activity help: baseline help added	0.46	−1.59	4.21
	(2.35)	(2.44)	(4.82)
Rate, no new limitation (%)		4.09	
Stricter inadequacy: baseline help added	2.52	−3.53	13.61**
	(3.00)	(2.99)	(6.59)
Rate, no new limitation (%)		7.17	
Observations		1,423	
Caregivers with new limitations	104	63	41

Notes: Adjusted percentage-point differences relative to zero new limitations, with couple-bootstrap standard errors below estimates. Stars use unadjusted two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Rate rows report the overlap-weighted percentage among observations whose caregiver develops no new limitation. All rows use the same observations. Main controls are those in equation (3). The remaining rows add baseline adequacy categories to both models and refit the weights. The main-controls binary and severity estimates each have 998 successful draws out of 999. With baseline help included, there are 997 successful binary draws and 998 severity draws. The any-limitation inadequacy comparison in the first row has a Holm-adjusted p-value of 0.156 in the two-outcome family comprising this outcome and absence of a personal-care provider. A supplementary six-test family includes both outcomes for any, one, and multiple new limitations versus zero; the multiple-limitations inadequacy comparison in the first row has adjusted p=0.0006. Table 6 uses the 1,381 observations that also have valid household-support answers. On that sample, it adjusts for baseline adequacy and divides the first outcome into no activity help and help received but not always sufficient; its estimates therefore differ slightly from the corresponding rows here.

tive counts locate the reports within interview modes; the adjusted estimates in Table D.1 retain all valid answering modes.

D.2 Baseline support users and follow-up selection

Table D.2 records survival, residence, and couple linkage before support-response restrictions. The support-loss comparisons below then follow recipients who received outside help at baseline.

Table D.2: Selection into the intact-community-couple population

	Zero limitations	Multiple new limitations
Caregiver health classified	4,245	197
Intact community couple	3,146	129
Excluded from that population	1,099	68
Excluded share (%)	25.89	34.52
Exclusive reasons for exclusion		
Recipient death	798	55
Recipient current institution	41	7
Other relationship or residence condition	260	6
Both household support channels observed	2,317	89

Notes: Counts use the expanded baseline risk set and require an observed zero-versus-multiple caregiver ADL classification. Death is assigned first, institution second, and all remaining exclusions third. Institution denotes recorded current residence, not necessarily a new admission. Caregiver death and unknown caregiver ADL are not assigned to either exposure group. The final row additionally requires both support measures. These are unweighted observation counts, not regression estimates.

D.3 Does the original recipient lose outside support?

Table D.3 follows recipients receiving help from children or home services at baseline. Loss is valid receipt at baseline followed by a valid zero at follow-up. Complete loss requires at least one baseline channel and valid information on both channels at both dates. Each baseline-user sample has its own fitted propensity and outcome adjustment. The two channel-specific losses form a family adjusted using the procedure of Holm (1979); complete loss and personal-care loss are secondary outcomes.

Among baseline home-service users in intact community couples, the adjusted loss difference is -17.8 percentage points (s.e. 10.8; Holm-adjusted $p = 0.197$). Four of the 19 multiple-new-limitation observations lose that channel, two with recipient ADL recovery. Complete loss occurs in three of the 20 multiple-new-limitation observations, one with recovery. The child-help sample has 12 multiple-new-limitation observations and does not meet the no-exclusion estimation rule; the table reports its sample size but no coefficient.

Pooling all new ADL limitations gives secondary loss differences of -17.8 points for help from children (s.e. 13.8), -12.8 for home services (s.e. 7.7), and -16.7 for all measured support (s.e. 5.7). Follow-up recovery is described but is not included in the regression adjustment. The next comparison extends the analysis beyond intact community couples.

Table D.3: Loss of the original recipient's baseline outside support

	Help from children	Home services	All measured support
Adjusted loss difference	–	-17.84^{*}	-21.48^{***}
		(10.80)	(8.01)
Loss rate, no new limitation (%)	–	43.54	38.10
Baseline-user observations	265	584	616
New-limitation observations	12	19	20
Successful bootstrap draws	–	350	321

Notes: Estimates are percentage points for multiple new limitations versus none, with couple-bootstrap standard errors below coefficients. Loss is receipt at baseline followed by no receipt in the channel. All measured support means loss of both help from children and home services among recipients initially receiving at least one, with both states observed at both interviews. The latter is a declared secondary outcome. The loss rate is the overlap-weighted percentage among observations whose caregiver develops no new limitation. The two channel-specific losses are primary; the service coefficient's two-test Holm-adjusted p-value is 0.197, retaining the unestimable child test in the family. Stars are unadjusted two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The multiple-new-limitation model for help from children discards 26 observations through complete prediction and fails the no-exclusion rule; its estimate is not reported. Home personal-care loss has 418 observations and 14 multiple-new-limitation transitions, but only 281/399 successful draws, below the 80 percent threshold, so inference is not reported. These outcomes concern observed support to the intact community couple, not deaths, institutional transitions, or unknown support.

D.4 Institutional transitions and unknown support

The selection analysis fixes the population using baseline support receipt and an observed zero-versus-multiple caregiver ADL classification. With both baseline channels valid and at least one present, the sample contains 1,377 observations, including 63 with multiple new limitations. Of those 63, 39 remain intact community couples, 18 record recipient death, and two institutional residence. Among the 1,314 zero-limitations observations, the corresponding counts are 923, 281, and 20.

Overlap weights are fitted on this full baseline-user population, including baseline support configuration. A valid community recipient's support is retained even if the couple no longer co-resides. One positive channel establishes continued support; two negatives establish complete loss. The first scenario treats institutional residence as termination of the community arrangement, leaving deaths and other unresolved states unknown. The second also counts recipient death as termination. With these weights, the conditional complete-loss difference among intact couples is -14.0 points. Including recorded institutions and known community recipients outside the original dyad gives -10.2 points (percentile interval $[-25.7, 12.9]$). Both conditional comparisons omit unresolved outcomes.

Table D.4 gives worst-case identification ranges. Positive weights and full baseline-group denominators allow every unresolved outcome to lie between zero and one. The lowest

possible difference assigns unresolved treated outcomes to zero and unresolved controls to one; the highest does the reverse. No missing-at-random or monotone-selection assumption is imposed. These bounds concern weighted mean differences, not bounds on the adjusted regression coefficient, whose implicit weights need not be positive.

Table D.4: Baseline-user identification ranges allowing unknown follow-up

	Help from children	Home services	Either channel
Baseline-user observations	665	1,141	1,377
New-limitation observations	43	54	63
Institution-status scenario			
Lower bound (percentage points)	−62.35	−56.14	−60.27
Upper bound (percentage points)	67.69	49.02	53.13
Death-inclusive termination scenario			
Lower bound (percentage points)	−32.76	−19.06	−32.14
Upper bound (percentage points)	40.61	23.21	29.52
Successful bootstrap draws	398	395	398

Notes: Entries are endpoints of identification ranges, not confidence limits or regression coefficients. Institution denotes follow-up residence; death-inclusive termination is a different composite outcome. The first scenario leaves deaths unknown. Either channel requires both baseline channels valid and at least one positive. Its propensity includes baseline support configuration. All samples are fixed before follow-up observation restrictions; each bootstrap draw refits the propensity score. The institution-scenario endpoints for either channel have percentile intervals [−68.6, −49.5] and [40.9, 67.1] points; the terminal-scenario endpoints have intervals [−43.0, −18.0] and [16.4, 43.9]. These intervals describe uncertainty about each bound endpoint.

Unresolved outcomes account for 58.5 percent of treated overlap weight and 54.9 percent of control weight in the institution scenario. Restricting baseline users to known community residents gives 1,305 observations and 60 multiple-new-limitation observations; the institution and death-inclusive ranges are [−62.3, 52.6] and [−33.2, 28.1], with 399 successful draws. In every specification, the ranges include both increased and reduced support loss.

E Gender and Regional Comparisons

The comparisons use multiple new caregiver limitations versus zero and fit a pooled propensity within each outcome sample, excluding one-new-limitation observations. Weighted outcome regressions interact exposure with caregiver gender or region and retain the baseline adjustment variables. Subgroup coefficients describe differences from the zero-limitations group; interaction tests compare those coefficients across groups. The outcome families cover household receipt, additional household measures, role entry, additional entry patterns, role levels, care transitions, and adequacy measures. Benjamini–Hochberg adjustment is applied

within these families and globally across 134 interaction tests per estimator (Benjamini and Hochberg, 1995).

E.1 Gender and the continuation of partner care

Table 7 reports household receipt and individual entry by caregiver gender. In the same pooled model, receipt of neither channel falls by 23.7 percentage points for male caregivers (s.e. 6.3) and 19.6 points for female caregivers (s.e. 5.9); none of the household interaction tests passes the household-family adjustment. Table E.1 decomposes exit into handoff, no measured provider, recovery, and an unresolved provider. The male-minus-female exit contrast is 23.3 percentage points (s.e. 9.0); its interaction test has family-adjusted $q = 0.043$ and global $q = 0.105$. Most couples are opposite-sex, so caregiver-gender comparisons also largely distinguish the recipient's gender.

Table E.1: Destinations of partner-care exit by caregiver gender

	Male caregiver	Female caregiver
Partner-care exit	32.89***	9.57
	(6.12)	(6.96)
Nonpartner handoff	−0.66	2.02
	(3.17)	(4.00)
No measured provider	1.55	−2.10
	(3.72)	(2.05)
Recipient recovery	9.57	−2.61
	(7.94)	(5.32)
Unresolved provider after exit	22.43***	12.26**
	(6.92)	(5.94)
New-limitation observations	49	55

Notes: All estimates are in percentage points for multiple new limitations versus none. Bootstrap standard errors are in parentheses on the following line. Stars use two-sided normal-approximation tests based on those standard errors: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A pooled gender-interaction model uses 2,445 observations; the first row is the exit estimate in Panel A of Table 7. Exit equals the sum of handoff, no measured provider, recovery, and unresolved-provider components. Recovery and unresolved provider states account for much of the exit difference; exit is not an unmet-need measure.

E.2 Regions, initial service use, and country composition

Northern and Western Europe comprises Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Luxembourg, the Netherlands, Sweden, and Switzerland. Southern Europe

comprises Cyprus, Greece, Italy, Malta, Portugal, and Spain. Central and Eastern Europe comprises Bulgaria, Croatia, the Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Slovakia, and Slovenia. Contributing countries vary by outcome sample.

The exit sample contains 44, 32, and 28 observations with multiple limitations in these regions, the household sample 36, 26, and 27, and the personal-care entry sample 22, 10, and 19. The joint test of equal regional exit coefficients has $\chi^2(2) = 13.57$ (unadjusted $p = 0.001$), family-adjusted $q = 0.038$, and global $q = 0.051$. The regional interaction for caregiver home-personal-care entry has family-adjusted $q = 0.010$ and global $q = 0.046$. Household, recipient-entry, and paired-role interactions do not pass the corresponding family adjustment.

Table E.2 reports initial personal-care use, within-region balance, and country sensitivity. Adding exposure interacted with recipient baseline personal-care receipt gives a Central/East-minus-North/West caregiver-entry contrast of -32.4 points at initial recipient receipt equal to zero. The contrast ranges from -39.5 to -30.8 when each of the 26 contributing countries is excluded in turn. All 24 exclusions meeting the bootstrap threshold have percentile intervals below zero. The Southern estimate is sensitive to Spain, which contributes eight of ten multiple-limitations observations.

Table E.2: Starting support and country sensitivity of caregiver service entry (multiple new limitations)

	Northern and Western	Southern	Central and Eastern
Baseline home personal-care receipt (percent)			
Caregiver, zero-limitation group	0.79	1.81	0.18
Caregiver, multiple-new-limitations group	9.09	10.00	0.00
Recipient, zero-limitation group	28.27	20.18	17.48
Recipient, multiple-new-limitations group	40.91	30.00	10.53
Observations	779	451	574
New-limitation observations	22	10	19
Multiple-new-limitation effective sample size	21.79	9.53	17.79
Maximum within-region SMD	0.506	0.823	0.558
Caregiver entry contrast (percentage points)			
Recipient baseline receipt fixed at zero	31.77***	29.11**	−0.58
	(12.03)	(14.09)	(4.15)
Leave-one-country-out point range	[33.2, 41.1]	[−5.8, 47.7]	[0.5, 3.5]

Notes: Baseline receipt percentages are unweighted and precede the caregiver transition. ESS and standardized mean differences (SMD) use the pooled propensity weights; the maximum checks health, demographic and country terms within region. Conditional entry coefficients come from a pooled outcome model adding treatment by recipient baseline personal-care receipt, evaluated at receipt equal to zero. They are not population-standardized marginal effects. Couple-bootstrap standard errors are below those coefficients; stars use two-sided normal-approximation tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. There are 376 successful draws out of 399 in this model. Point-estimate ranges cover all 26 country exclusions. Spain and Belgium exclusions have 274 and 247 successful draws, below the 80 percent threshold; their confidence intervals are withheld. All other exclusions meet the rule. The Southern exposure group includes eight Spanish and two Italian observations. These are sensitivity checks of the existing regional finding.

F Second-Follow-up Comparisons and Support Transitions

F.1 Scheduling, exposure, and outcome definitions

A second interview is scheduled using country participation, with 5,269 opportunities and 1,090 baseline observations without a later participating wave. Caregiver health groups remain fixed at the first follow-up. Among the scheduled observations, 3,469 caregivers initially have zero limitations, 248 have one, and 171 have multiple. The propensity and outcome regressions include the scheduled gap between follow-ups.

The common-sample regressions measure receipt at each follow-up, including support already present at baseline. They therefore require reports at both follow-ups rather than

baseline nonreceipt. Household outcomes require both channels; paired service outcomes require both original partners. Valid regular first-follow-up exit reports in wave 7 are retained. Table F.1 reports the full comparisons underlying Section 5.1.

Table F.1: Care and outside support at two follow-ups

	First follow-up	Second follow-up	Second minus first	Observations (new limitations)
A. Partner-care exit				
Any new limitation versus none	11.8**	12.8**	0.9	988 (91)
	(5.6)	(5.2)	(6.0)	
Multiple new limitations versus none	9.2	12.9*	3.7	930 (33)
	(8.5)	(7.6)	(10.7)	
B. Household receipt: any new limitation versus none				
Help from children	10.9	6.6	−4.4	689 (57)
	(7.3)	(6.9)	(8.4)	
Home services	8.3	3.0	−5.3	689 (57)
	(6.7)	(6.6)	(6.7)	
Both channels	10.8*	9.7	−1.0	689 (57)
	(6.0)	(6.5)	(6.8)	
C. Individual receipt: any new limitation versus none				
Original caregiver: home services	16.7***	4.3	−12.4**	712 (62)
	(5.7)	(5.7)	(5.9)	
Original recipient: home services	0.8	−0.1	−0.9	712 (62)
	(5.9)	(6.2)	(7.3)	
Original caregiver: home personal care	15.4***	12.6***	−2.8	712 (62)
	(4.5)	(4.6)	(4.7)	
Original recipient: home personal care	−0.1	0.4	0.6	712 (62)
	(5.1)	(5.9)	(6.6)	

Notes: Adjusted percentage-point differences, with couple-bootstrap standard errors below estimates. Caregiver health groups are fixed at the first follow-up. Each row uses the same observations, overlap weights, and regressors at both interviews; changes are estimated directly. Household outcomes require valid information on both channels at both dates. Individual service outcomes require valid reports for both original partners at both dates. Models use baseline health and demographic controls, country effects, and the scheduled first-to-second survey gap. They do not condition on first-follow-up support receipt. There are 998 successful draws for any-limitation exit, 873 for multiple-limitation exit, 987 for household receipt, and 988 for individual receipt, out of 999. The exit samples retain valid regular first-follow-up reports in wave 7, as in the earlier two-follow-up comparison. Individual child-help receipt is examined through Table F.3; the primary-calendar propensity model for that smaller common sample does not retain all observations. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F.2 Estimation coverage and changing health

Table F.2 reports estimation coverage and successful draws from 999 couple-level bootstrap resamples. The any-new-limitation household and individual-service models meet the 80 percent success threshold. The corresponding multiple-limitation support models use smaller samples and fall below that threshold. Individual child-help point fits exclude six obser-

vations through complete prediction; their support transitions are reported descriptively instead.

Table F.2: Common-sample second-follow-up estimation coverage

Outcome	Observations (new limitations)	Zero-group ESS	New-limitations ESS	Successful draws
Any new limitation versus none				
Partner-care exit	988 (91)	564	89	998/999
Household receipt	689 (57)	381	56	987/999
Individual child-help receipt	336 (20)	175	20	–
Individual service receipt	712 (62)	408	61	988/999
Individual personal-care receipt	712 (62)	408	61	988/999
Multiple new limitations versus none				
Partner-care exit	930 (33)	426	33	873/999
Household receipt	651 (19)	291	19	586/999
Individual child-help receipt	320 (4)	36	4	–
Individual service receipt	670 (20)	275	20	593/999
Individual personal-care receipt	670 (20)	275	20	593/999

Notes: Diagnostics use the scheduled country calendar and 999 requested couple resamples. ESS denotes effective sample size. Household receipt requires both channels at both interviews; individual service and personal-care rows require reports for both partners. Child-help counts describe the retained point fit, which excludes six specified observations; no coefficient or bootstrap inference is reported for that fit. Other point fits exclude no observations. Bootstrap inference requires at least 80 percent successful draws. Standard errors for household and individual-service comparisons under multiple limitations are therefore not reported.

Among the 62 caregivers with any new limitation in the paired service sample, 31 report zero ADL limitations at the second interview; 49 of the 650 initially zero-limitations caregivers report later difficulties. Across all 171 initially multiple-limitations observations with a second opportunity, later caregiver states are zero limitations (33), one (27), multiple (39), death (35), and unknown (37). Recipient death is recorded in 81 of these 171 observations, compared with 1,210 of 3,469 initially zero-limitations observations. These counts concern the scheduled population, not the common regression sample.

F.3 Later entry and receipt of newly added assistance

Table F.3 follows each original role separately. Panel A begins with nonreceipt at the first follow-up and identifies later entry. Panel B begins with new receipt between baseline and the first follow-up and records whether it is reported again. Individual follow-up can remain valid after the original partners stop living together. Death, institution, and unknown receipt are retained in the denominators.

Among original caregivers with known subsequent community receipt, 14 of 15 first entrants into help from children receive it again, as do 11 of 13 first entrants into home services.

Table F.3: Later entry and receipt of previously added support

	First-follow-up observations	State at the second follow-up				
		No receipt	Receipt	Death	Institution	Unknown
A. No receipt at the first follow-up						
Help from children: caregiver	81	18	7	10	1	45
Help from children: recipient	56	10	5	13	0	28
Home services: caregiver	180	63	14	23	1	79
Home services: recipient	103	32	8	20	2	41
Home personal care: caregiver	223	83	12	33	2	93
Home personal care: recipient	126	45	7	23	2	49
B. New receipt between baseline and the first follow-up						
Help from children: caregiver	44	1	14	4	0	25
Help from children: recipient	19	0	3	3	1	12
Home services: caregiver	36	2	11	7	1	15
Home services: recipient	20	2	1	5	0	12
Home personal care: caregiver	30	4	5	5	0	16
Home personal care: recipient	18	1	2	6	0	9

Notes: Unweighted observation counts among original caregivers with any new ADL limitation at the first follow-up and their original recipients. Each row begins with a valid individual community report and a scheduled second interview. The five subsequent-state columns are mutually exclusive and sum to the first-follow-up count. Panel A identifies later entry; Panel B identifies assistance reported at both follow-ups after no receipt at baseline. Two positive interviews do not establish uninterrupted assistance between them. Individual receipt can remain observed after the original partnership ends, so these samples differ from the couple-complete regressions in Table 8. Home personal care is a component of home services; rows across channels are not additive. Caregiver and recipient denote fixed baseline roles.

The full first-entrant counts are 44 and 36. Later entry also occurs for both roles: among initial nonusers, home services reach 14 caregivers and eight recipients, while help from children reaches seven and five. These are observation counts rather than adjusted contrasts.

F.4 Partner care after a first exit

Among first exits with known later community care status, four of 21 multiple-limitations observations resume partner care (19.0 percent), compared with 128 of 429 zero-limitations observations (29.8 percent). The remaining 45 of 66 first exits in the multiple-limitations group comprise 17 recipient deaths, two institutional moves, and 26 unknown care states.

Table F.4: Second-interview states after first-interview partner-care exit

First-follow-up caregiver health	Zero limitations	Multiple new limitations
First exit with second scheduled opportunity	914	66
Exit still recorded	301	17
Partner care resumes	128	4
Recipient death	112	17
Recipient institution	10	2
Unknown later care	363	26

Notes: Unweighted observation counts, with health groups fixed at the first follow-up. Each column retains every first observed exit with a second scheduled country wave. Death and institution take precedence over community care. Continued exit denotes exit at both interviews, not an uninterrupted spell between them. Original roles and the baseline reporter anchor remain fixed.

G Timing Around Recorded Functional Change

The lead comparison requires a zero caregiver ADL count at the current interview and a valid count at the next scheduled country wave. Original roles stay fixed even if partner care has ended. Receipt models use current support indicators; change models additionally require baseline receipt and add it to both models. Prior caregiver ADL history enters outcome adjustment but is omitted from the diagnostic propensity because the prior-positive category contains only one future event in the exit sample.

The same-sample comparison requires outcomes at both dates. Exposure is the caregiver's ADL transition between them; within each outcome, observations, baseline covariates, and overlap weights are identical across dates. Every bootstrap draw re-estimates both stages and directly calculates the later-minus-earlier difference.

Table G.1: Future ADL limitations, current partner-care exit, and household receipt

	Adjusted difference
Current partner-care exit	9.66**
	(4.58)
Observations	1,377
Observations with a future limitation	121
Current household receipt: children	7.65
	(4.65)
Current household receipt: services	2.78
	(4.57)
Current household receipt: both	3.82
	(3.38)
Observations	1,330
Observations with a future limitation	119
Current change in help from children	9.79*
	(5.59)
Current change in services	8.29
	(5.99)
Observations	853
Observations with a future limitation	79

Notes: Coefficients are percentage-point differences. Couple-bootstrap standard errors are below estimates. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Each draw re-estimates both model stages; all reported estimates have 999 successful draws. Current caregiver ADL is zero. Future ADL is observed at the next scheduled country wave with a later interview date. No future eligible care relationship is required. Prior caregiver ADL history is omitted from the propensity model because its prior-positive category contains only one future event in the exit sample; it enters the outcome regression. Baseline household channel states additionally enter change models. Future observation selects this three-interview population.

Table G.2: Partner-care exit and household receipt in the same three-interview sample

	Earlier	Later	Later minus earlier
Partner-care exit	11.26*	3.22	−8.04
	(6.78)	(6.65)	(8.82)
Observations		745	
Observations with a future limitation		61	
Household receipt: children	−5.46	2.05	7.51
	(6.18)	(7.01)	(7.68)
Household receipt: services	−7.23	6.56	13.79*
	(6.58)	(6.76)	(7.80)
Household receipt: both channels	−8.64**	3.57	12.21*
	(3.86)	(6.11)	(6.88)
Observations		632	
Observations with a future limitation		47	

Notes: Estimates are percentage-point differences, with couple-bootstrap standard errors below coefficients. Stars use two-sided normal-approximation tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All models have 999 successful draws. The exposure, weights, and adjustment set are fixed across dates within each outcome sample; both stages are re-estimated in each draw. Prior caregiver ADL history enters outcome adjustment but not the diagnostic propensity, as in Table G.1. Of the 61 exit-sample transitions, 44 involve one new limitation and 17 involve multiple new limitations. Percentile intervals for all later-minus-earlier contrasts include zero.

H Additional Health Definitions and Flexible Risk Adjustment

H.1 Caregiver acute medical events

The acute-event comparison starts from the same 6,359 baseline care observations. Dedicated questions about events since the last interview identify heart attacks, strokes or cerebrovascular events, and hip fractures. Recurrence confirmations distinguish a new episode from an earlier condition; confirmed recurrences are included. The preceding completed main interview must coincide with the selected baseline. Reported dates are checked against the caregiver's interview interval where available. A valid no-event observation requires all three component answers to be negative. No subsequent ADL change is required. Baseline and earlier recorded histories of heart problems, stroke, and hip fracture enter both the propensity and outcome models.

The source classification contains 179 event observations, 3,650 no-event observations, and 2,530 with unresolved event status. The comparison reported here additionally requires resolved baseline histories, leaving 179 event and 3,647 no-event observations before outcome-specific eligibility. Controls can experience functional decline without one of these medical events. The outcomes and reporting rules are otherwise those of the main analysis. The bootstrap resamples couples and re-estimates the propensity and outcome models in every draw.

A sensitivity comparison requiring reported timing to place an acute component within the interview interval gives a joint-household-receipt difference of 10.97 points (s.e. 4.28) and caregiver service entry of 10.61 points (s.e. 4.90). Restricting instead to caregivers with no recorded baseline or earlier PH006 history gives an exit difference of 14.08 points (s.e. 6.06), based on 1,922 observations and 59 events, with 865 successful bootstrap draws. The recorded-history restriction and the event-date restriction define separate comparisons.

H.2 Task-weighted functional difficulty

Let D_j denote a reported difficulty in each of the six tasks, ordered as dressing, walking across a room, bathing, eating, getting in or out of bed, and toileting. The supplementary index is

$$B = 2D_{\text{dress}} + 3D_{\text{walk}} + D_{\text{bath}} + 2D_{\text{eat}} + 3D_{\text{bed}} + 2D_{\text{toilet}}. \tag{H.1}$$

The weights follow the relative maximum points for the corresponding activities in the standard Barthel index, divided by five (Mahoney and Barthel, 1965). SHARE records task

Table H.1: Acute caregiver events: partner care and outside support

Outcome	Adjusted difference	Observations	Events	Successful draws
Partner-care exit	6.86	2,558	117	978
	(4.98)			
Household receipt: help from children	10.41**	2,506	109	994
	(4.89)			
Household receipt: home services	6.04	2,506	109	994
	(4.56)			
Household receipt: both channels	9.17**	2,506	109	994
	(4.09)			
Entry into help from children: G	10.64*	1,505	58	976
	(5.63)			
Entry into help from children: R	13.22**	1,505	58	976
	(5.90)			
Paired child-help entry: G minus R	−2.58	1,505	58	976
	(5.47)			
Entry into home services: G	11.22**	1,860	79	992
	(4.70)			
Entry into home services: R	7.90*	1,860	79	992
	(4.53)			
Paired service entry: G minus R	3.32	1,860	79	992
	(5.00)			
Recipient help does not always meet needs	3.29	1,399	57	977
	(6.38)			

Notes: Differences are percentage points from overlap-weighted outcome regressions. G and R retain their original roles. Event observations have a newly reported medical episode, including confirmed recurrences; they are not selected on follow-up ADL. Each outcome uses its original eligibility rules. The adequacy sample contains community recipients with continuing ADL limitations. Standard errors below estimates come from 999 requested couple-bootstrap draws, with successful draws reported in the last column. Stars use unadjusted two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

difficulty rather than the complete Barthel assistance categories, and its walking and transfer questions have different definitions. I therefore use B as a six-item functional-difficulty index, with higher values indicating more weighted difficulty, rather than apply clinical Barthel cutoffs.

All original baseline records, item-validity rules, and outcome definitions are retained. Baseline caregiver B is zero. Any new ADL limitation is exactly equivalent to follow-up $B > 0$, so changing caregiver item weights alone leaves the main binary comparison unchanged. Replacing the recipient's baseline ADL count with baseline B in both the propensity and outcome regressions changes the any-new-limitation estimates for exit, household

joint receipt, and both partners' entry into help from children and home services by at most 0.1 percentage point, using the same 999 requested couple resamples.

The prespecified severity partitions are $B = 0$, 1–2, and at least 3, with a threshold of 4 as a sensitivity comparison. A score of at least 3 includes every multiple-item pattern and a single walking or bed-transfer difficulty. In the exit sample, this moves 19 one-item observations into the higher group, expanding it from the original 104 multiple-limitations observations to 123. A score of at least 4 requires more than one difficulty and contains 76 exit observations. The score thresholds are analytic definitions, not clinical diagnoses.

Table H.2: Partner-care exit under alternative severity measures

Measure and groups	Lower group versus zero	Higher group versus zero	Higher minus lower
Original count: one / multiple	−2.86	20.42***	23.28***
	(4.18)	(4.65)	(5.88)
Weighted score: 1–2 / at least 3	−1.77	14.95***	16.73***
	(4.74)	(4.45)	(6.25)
Weighted score: 1–3 / at least 4	0.86	19.85***	18.99***
	(3.96)	(5.51)	(6.57)
Observations		2,581	
Zero-limitations observations		2,341	
Lower / higher group: original count		136 / 104	
Lower / higher group: threshold 3		117 / 123	
Lower / higher group: threshold 4		164 / 76	

Notes: All estimates retain any-onset overlap weighting, rather than fit separate high-versus-zero propensities. The original-count row uses the same primary-analysis bootstrap results as Table 4, including the directly estimated multiple-minus-one contrast. It uses the original recipient baseline ADL count; weighted-score rows use the recipient's baseline weighted index in both stages. Keeping the original baseline covariate gives high-group exit estimates of 14.97 and 19.81 points for the two weighted thresholds. Every row has 999 successful couple-bootstrap draws; the weighted-score rows use the alternative-index resamples. Stars use two-sided normal tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All thresholds and contrasts were specified before estimating these results.

With the weighted baseline adjustment, household joint receipt is 20.89 points more common in the score-at-least-3 group (s.e. 5.06) and 23.20 points more common in the score-at-least-4 group (s.e. 6.38). Caregiver child entry differs by 17.63 and 21.85 points, respectively (s.e. 6.19 and 8.95); caregiver service entry differs by 15.64 and 18.26 points (s.e. 5.02 and 6.84). Continuous count and weighted-score regressions, scaled by one full-sample standard deviation, give similar directions and magnitudes: exit differs by 4.38 and 4.10 points, joint receipt by 4.00 and 3.86, caregiver child entry by 3.33 and 2.85, and caregiver service entry by 3.25 and 2.79.

The replication files also include joint six-item regressions, without simultaneously in-

cluding their total count or weighted sum, and prespecified bootstrap-covariance tests of equal item coefficients and Barthel-proportional coefficients. Neither restriction is rejected for exit at five percent; the eight selected outcome-by-restriction tests per baseline specification do not reject after Holm adjustment. The weighted comparisons preserve the main severity pattern.

H.3 External risk learning and target-sample residualization

The source data use country-scheduled person-intervals with baseline ADL zero, age at least 50, and a later survey opportunity. Both members of every target couple, 9,716 people in total, are excluded from source training. The risk models use age, gender, the common seven-item IADL measure, baseline diagnoses and other health measures, and earlier health levels and changes. They do not use follow-up health or care as predictors. Imputation, scaling, and encoding are fitted within training folds. Repeated people and linked partners remain together in the three source folds. The source ensemble averages regularized logistic regression, a random forest, and histogram gradient boosting; the external predictions for target caregivers average predictions from the three source-fold fits.

The acute-risk training sample contains 170,320 resolved person-intervals and 5,585 events, including confirmed recurrences. Its grouped out-of-fold AUC is 0.730; the AUC in the target observations with classified exposure is 0.702. Mean predicted and observed target risks are 4.29 and 4.68 percent. For any new ADL limitation, 212,094 source intervals contain 13,313 events, with source and target AUCs of 0.820 and 0.773, and mean predicted and observed target risks of 9.22 and 10.10 percent. The target sample includes a small number of caregivers younger than 50; their source-model predictions are retained, and risk adjustment is recalibrated within each target analysis sample.

Within each target outcome sample, I cross-fit both the treatment probability $m(X) = \Pr(D = 1 \mid X)$ and the outcome mean $\ell(X) = \mathbb{E}[Y \mid X]$. The baseline features include the external risks, caregiver health histories, recipient characteristics, and the baseline reporting and support variables used for that outcome. With $\widehat{V}_i = D_i - \widehat{m}_i$, the partialling-out estimate and influence score are

$$\widehat{\theta} = \frac{\sum_i \widehat{V}_i (Y_i - \widehat{\ell}_i)}{\sum_i \widehat{V}_i^2}, \tag{H.2}$$

$$\widehat{\psi}_i = \frac{\widehat{V}_i \{Y_i - \widehat{\ell}_i - \widehat{\theta}\widehat{V}_i\}}{n^{-1} \sum_j \widehat{V}_j^2}. \tag{H.3}$$

Both nuisance functions are fitted outside the observation's grouped target fold, following

Chernozhukov et al. (2018). I use three folds repeated three times and average the held-out predictions. The main target learner averages regularized logistic regression and histogram gradient boosting; a logistic-only specification is retained as a transparent alternative. No observations are trimmed. Estimated probabilities are bounded at 0.002 and 0.998; none of the final ensemble treatment predictions reaches these bounds.

Grouped target folds and inference keep observations linked by the same caregiver or original couple together. The conditional-variance weighting of the residualized comparison differs from a population-average contrast. Across the three target splits, the ADL joint-receipt estimates range from 11.96 to 12.13 points, caregiver child entry from 12.23 to 14.65, and caregiver service entry from 11.26 to 12.30. In the logistic-only version the corresponding estimates are 11.68, 13.54, and 11.70 points. The full set of predictions, calibration results, and repetition-specific estimates is retained in the replication files.

Table H.3: Flexible health-risk adjustment on the original outcome samples

Outcome	Any new ADL limitation	Acute medical event
Partner-care exit	5.45	6.13
	(3.42)	(4.60)
Household receipt: help from children	15.35***	13.01***
	(3.64)	(4.73)
Household receipt: home services	8.44**	5.16
	(3.42)	(4.37)
Household receipt: both channels	12.01***	9.78**
	(3.18)	(4.13)
Entry into help from children: G	13.43***	11.42**
	(4.21)	(5.33)
Entry into help from children: R	9.68**	14.36***
	(4.19)	(5.45)
Paired child-help entry: G minus R	3.75	−2.94
	(4.71)	(5.47)
Entry into home services: G	11.89***	11.49***
	(3.35)	(4.39)
Entry into home services: R	5.63*	9.09**
	(3.19)	(4.49)
Paired service entry: G minus R	6.26*	2.40
	(3.52)	(4.97)
Entry into home personal care: G	9.98***	7.54**
	(2.79)	(3.39)
Entry into home personal care: R	3.98	5.44
	(3.15)	(3.89)
Paired personal-care entry: G minus R	6.00	2.10
	(4.03)	(4.59)

Notes: Differences are percentage points. Standard errors below estimates use grouped influence scores and Student-t inference, not the main tables' bootstrap. Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. For ADL/acute comparisons, exit uses 2,581/2,558 observations with 240/117 exposed observations; household receipt uses 2,537/2,506 with 220/109; child entry uses 1,528/1,505 with 112/58; service and personal-care entry use 1,890/1,860 with 137/79. Each sample matches the corresponding conventional analysis. G–R contrasts use identical rows, treatment predictions, and paired outcome predictions. After Holm adjustment across 13 outcomes per exposure, ADL household child and joint receipt, and G entry into all three support measures, remain significant at five percent; the acute ensemble comparisons do not meet that familywise threshold.